

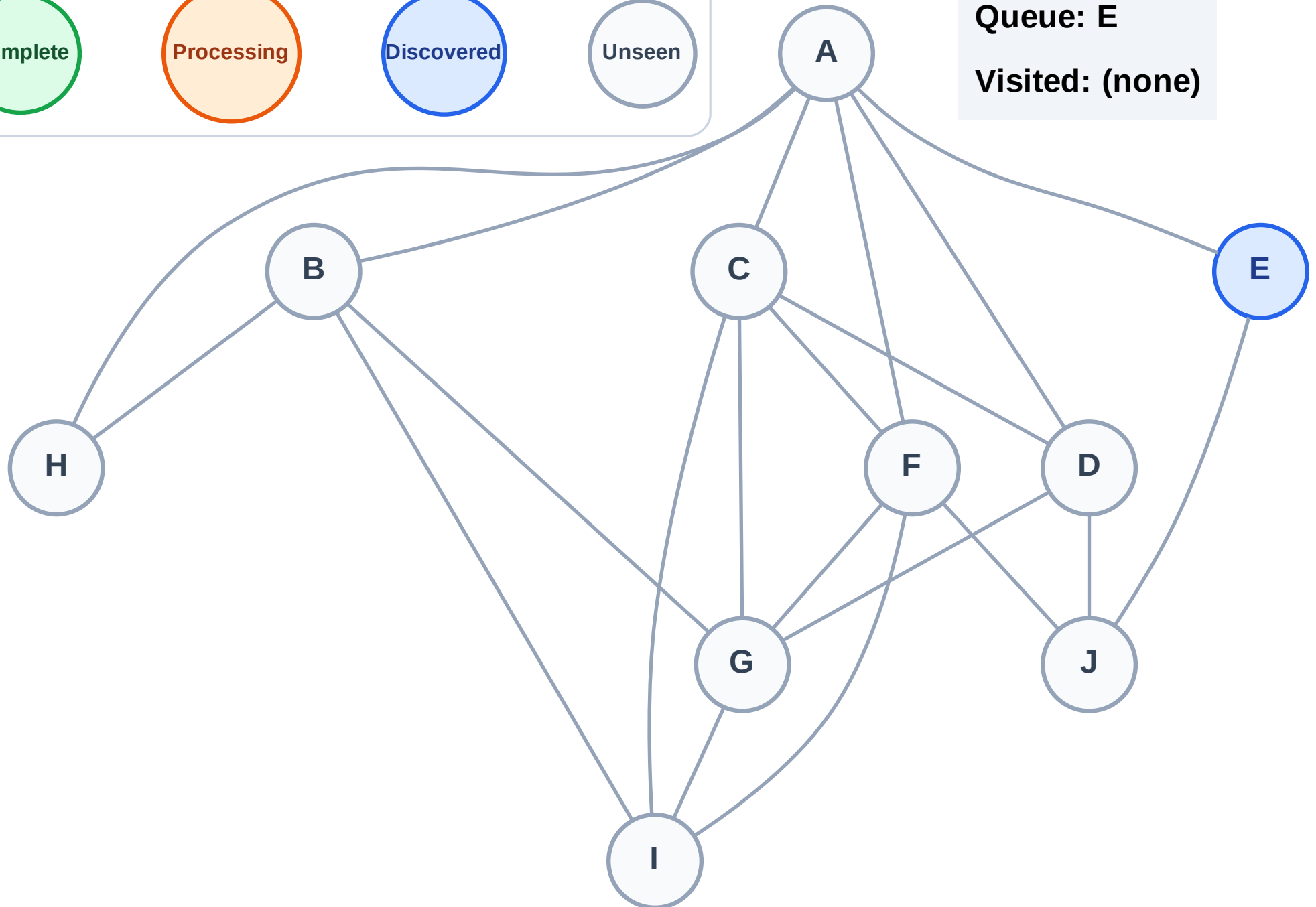
BFS Traversal

Step 1: Start at E

Legend



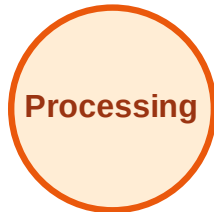
Queue: E
Visited: (none)



BFS Traversal

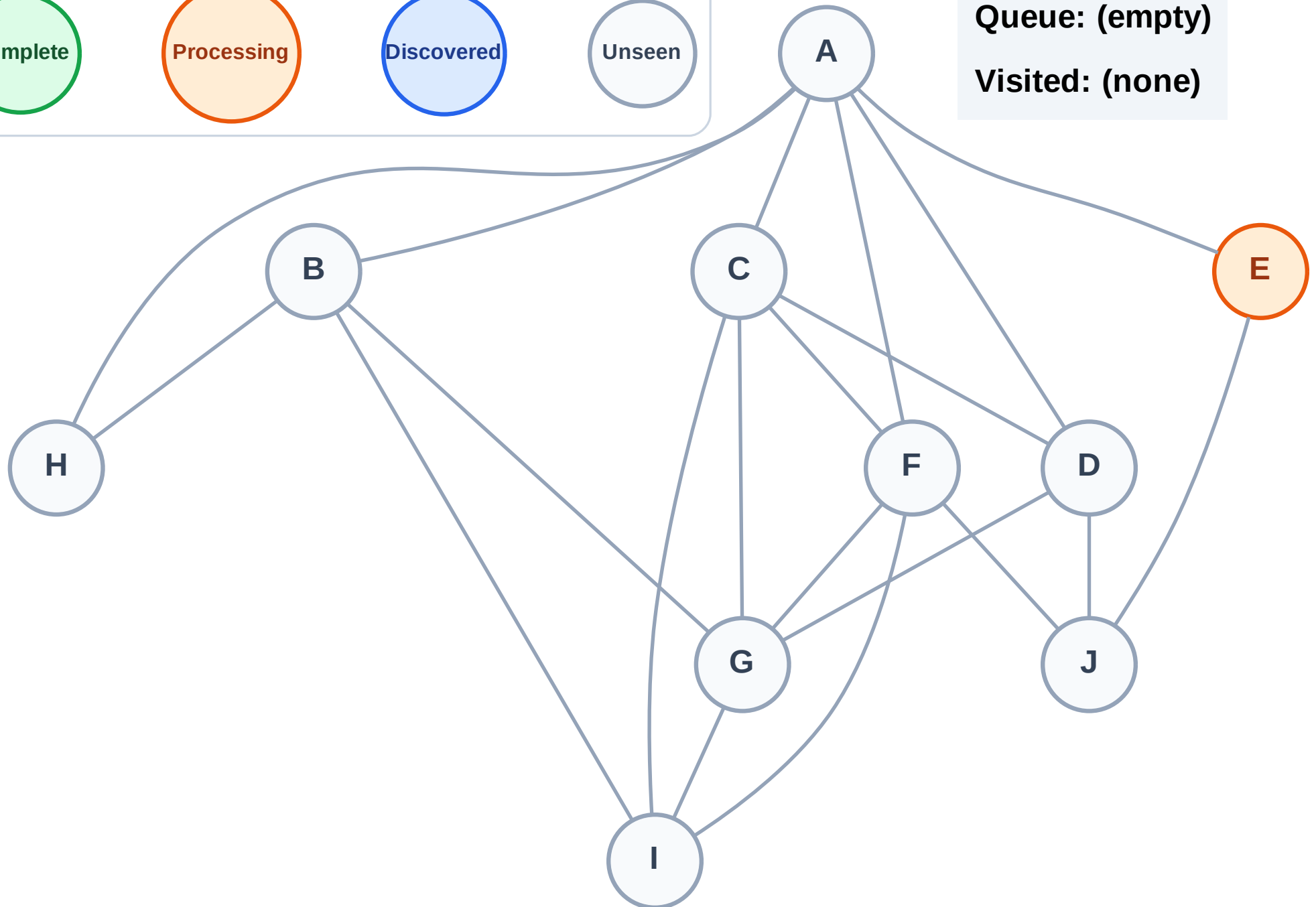
Step 2: Process node E

Legend



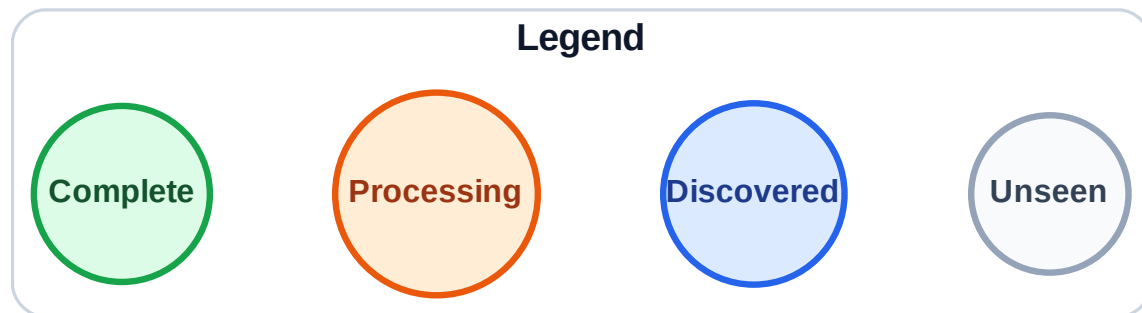
Queue: (empty)

Visited: (none)

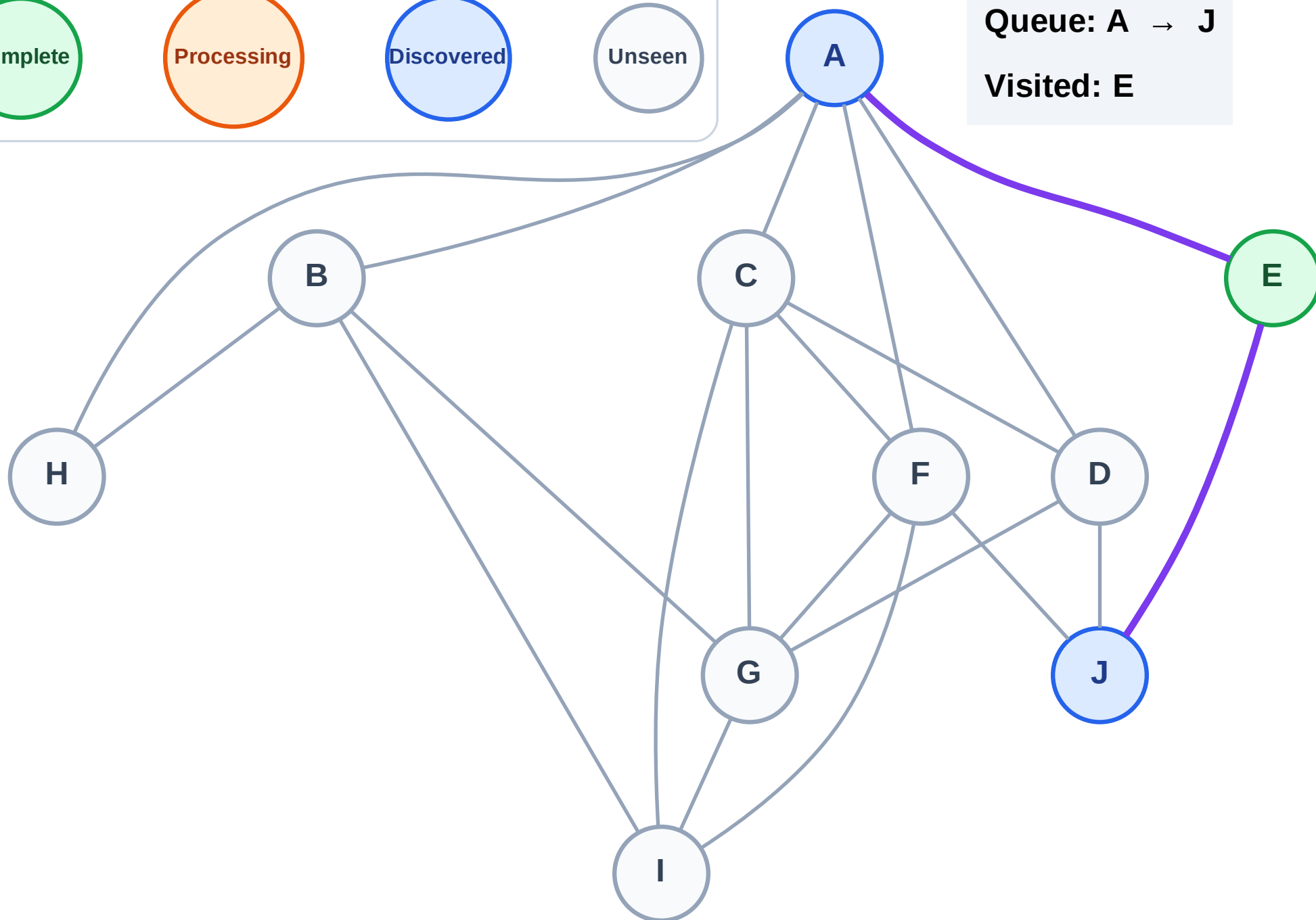


BFS Traversal

Step 3: Discover A, J from E



Queue: A → J
Visited: E



BFS Traversal

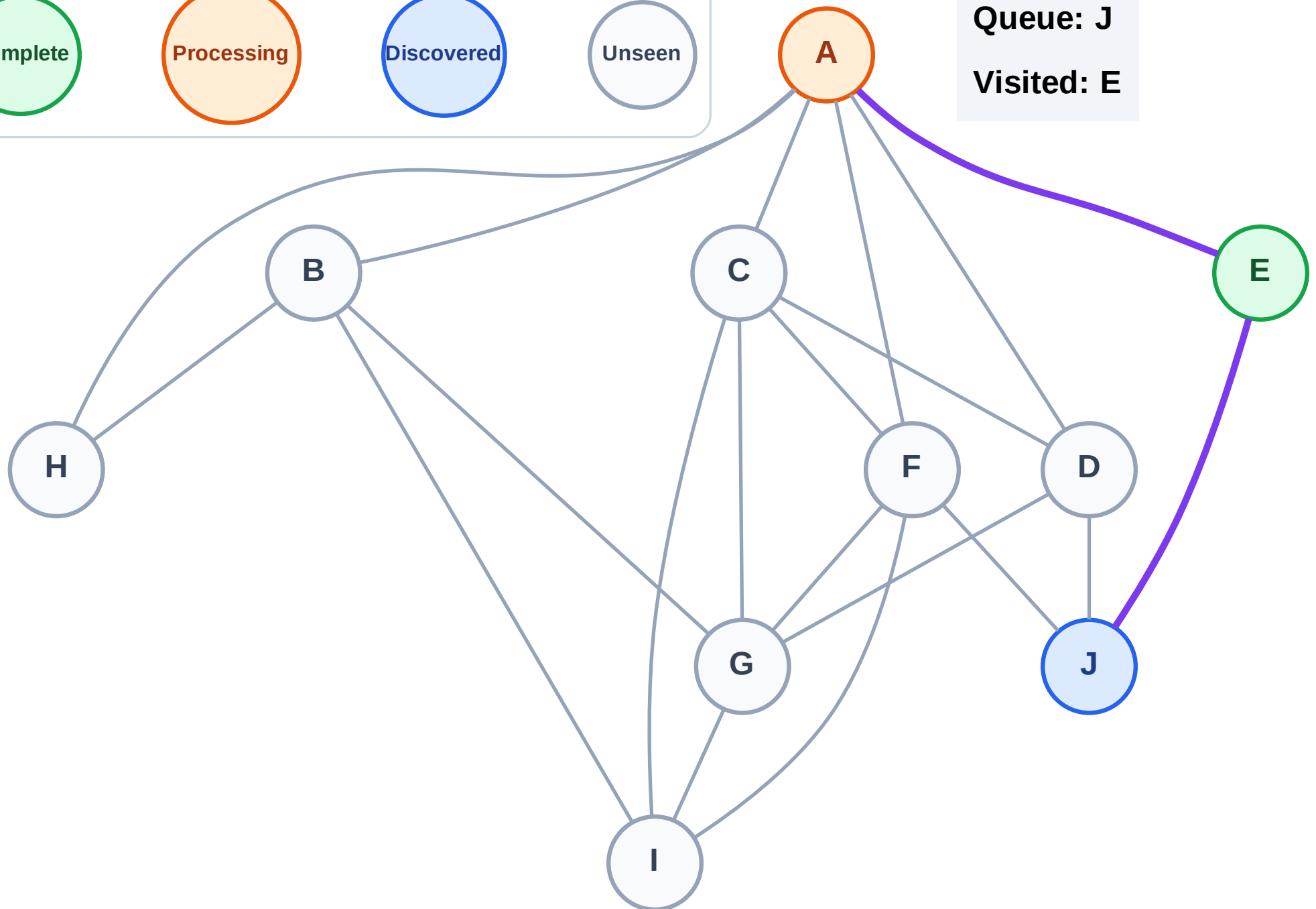
Step 4: Process node A

Legend



Queue: J

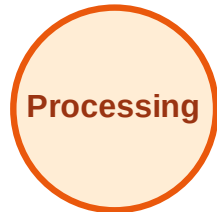
Visited: E



BFS Traversal

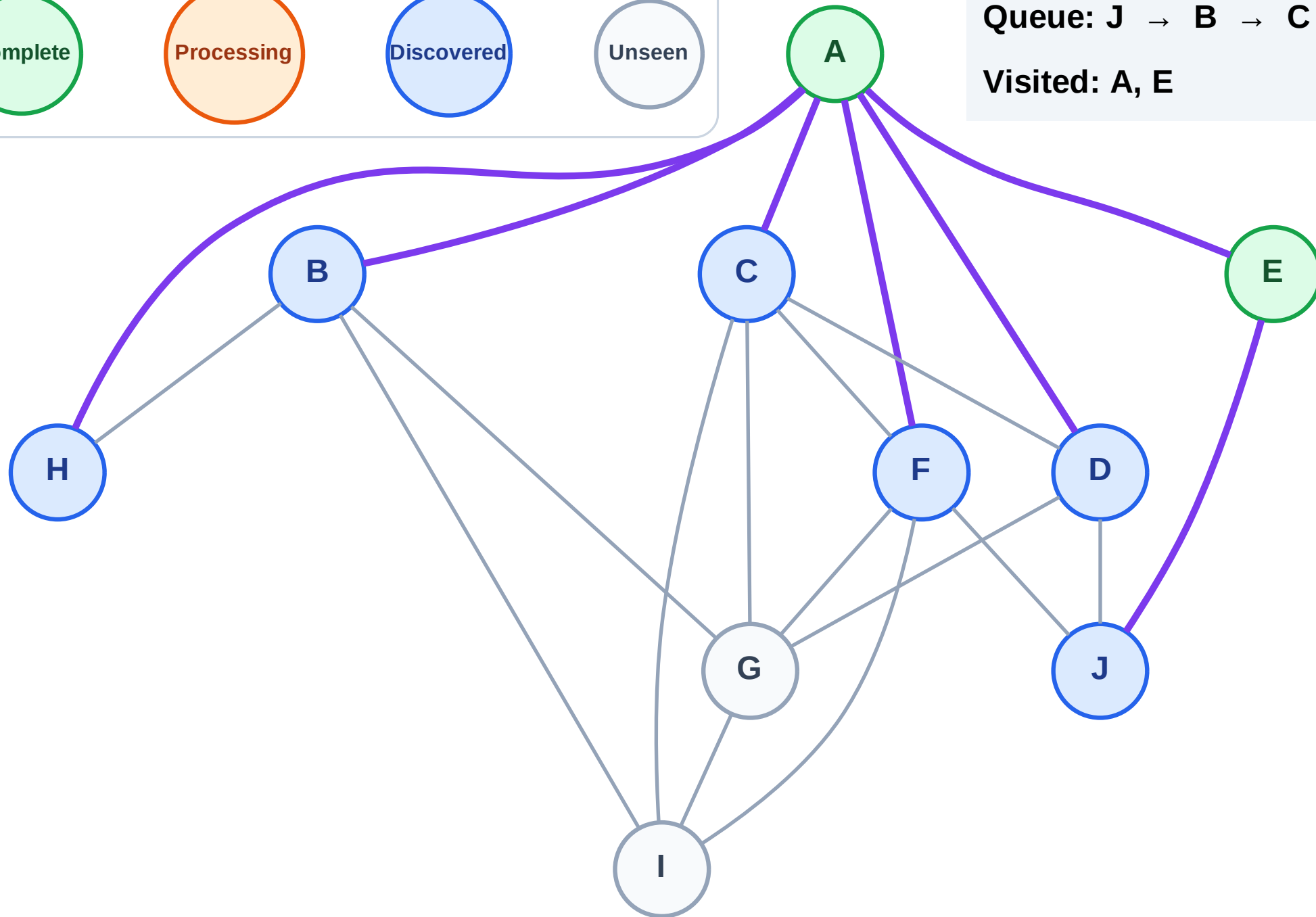
Step 5: Discover B, C, D, F, H from A

Legend



Queue: J → B → C → D → F → H

Visited: A, E



BFS Traversal

Step 6: Process node J

Legend

Complete

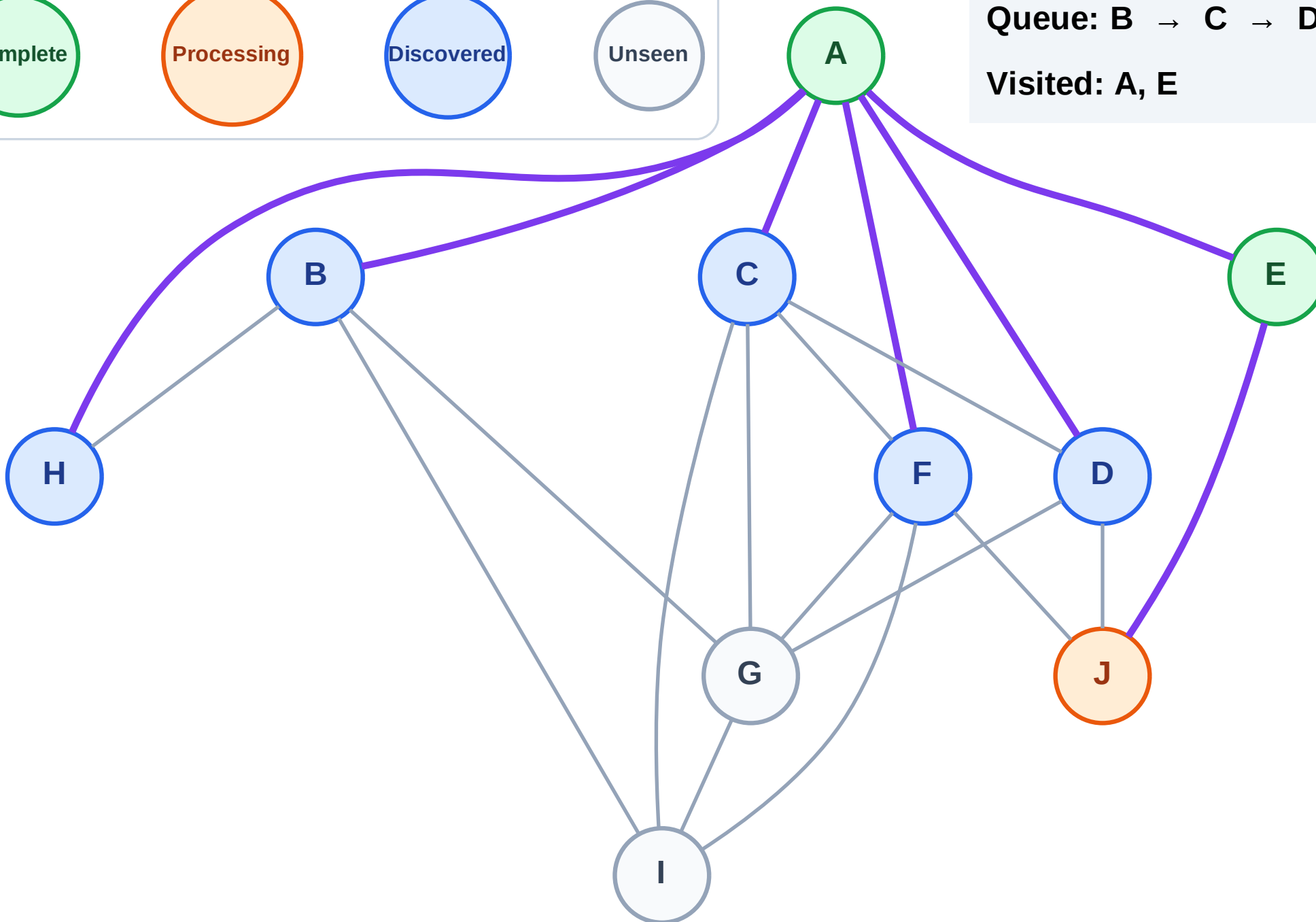
Processing

Discovered

Unseen

Queue: B → C → D → F → H

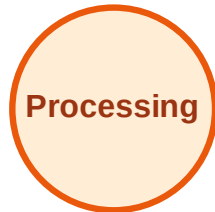
Visited: A, E



BFS Traversal

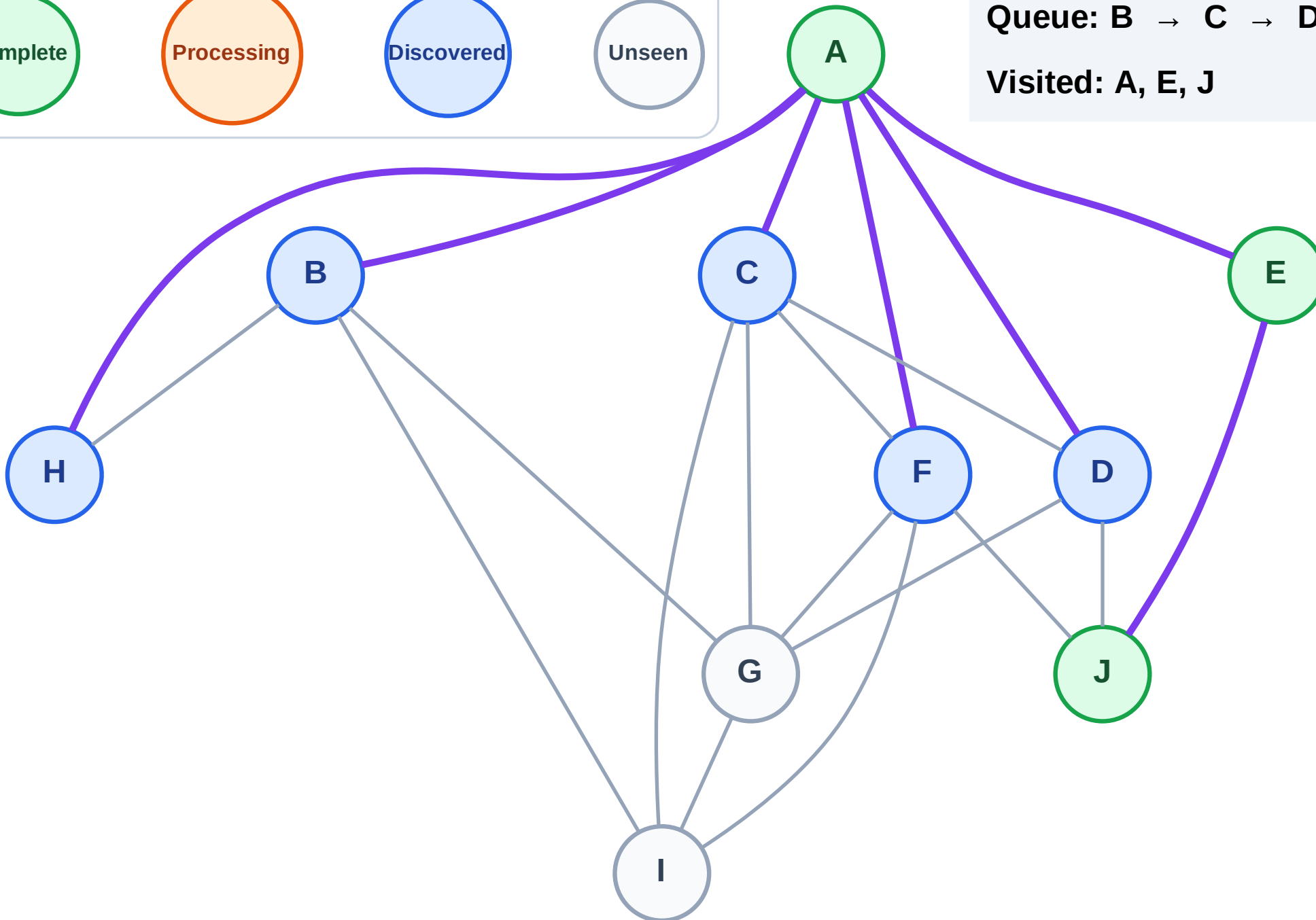
Step 7: No new neighbors from J

Legend



Queue: B → C → D → F → H

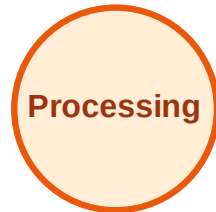
Visited: A, E, J



BFS Traversal

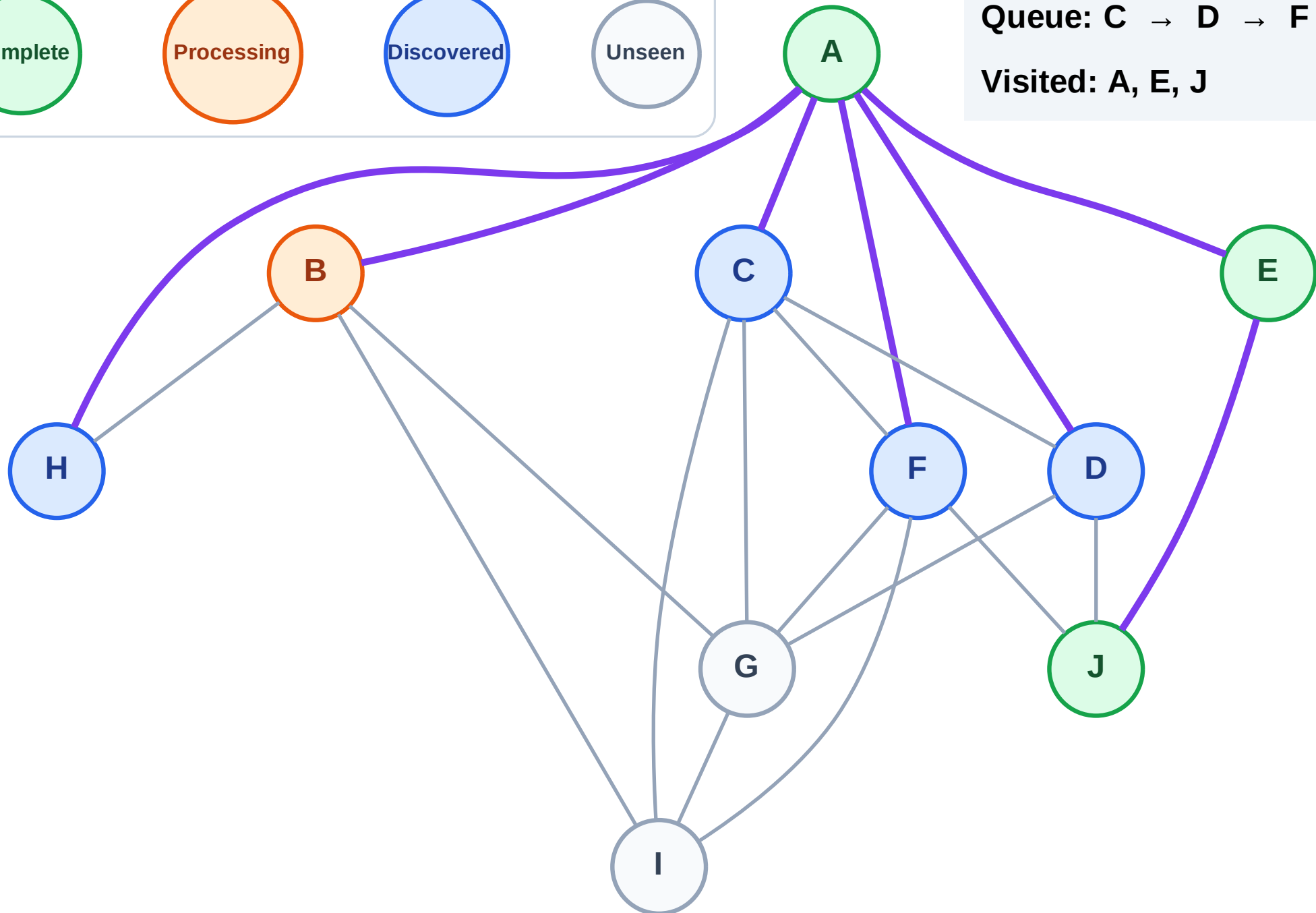
Step 8: Process node B

Legend



Queue: C → D → F → H

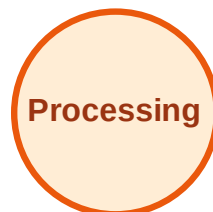
Visited: A, E, J



BFS Traversal

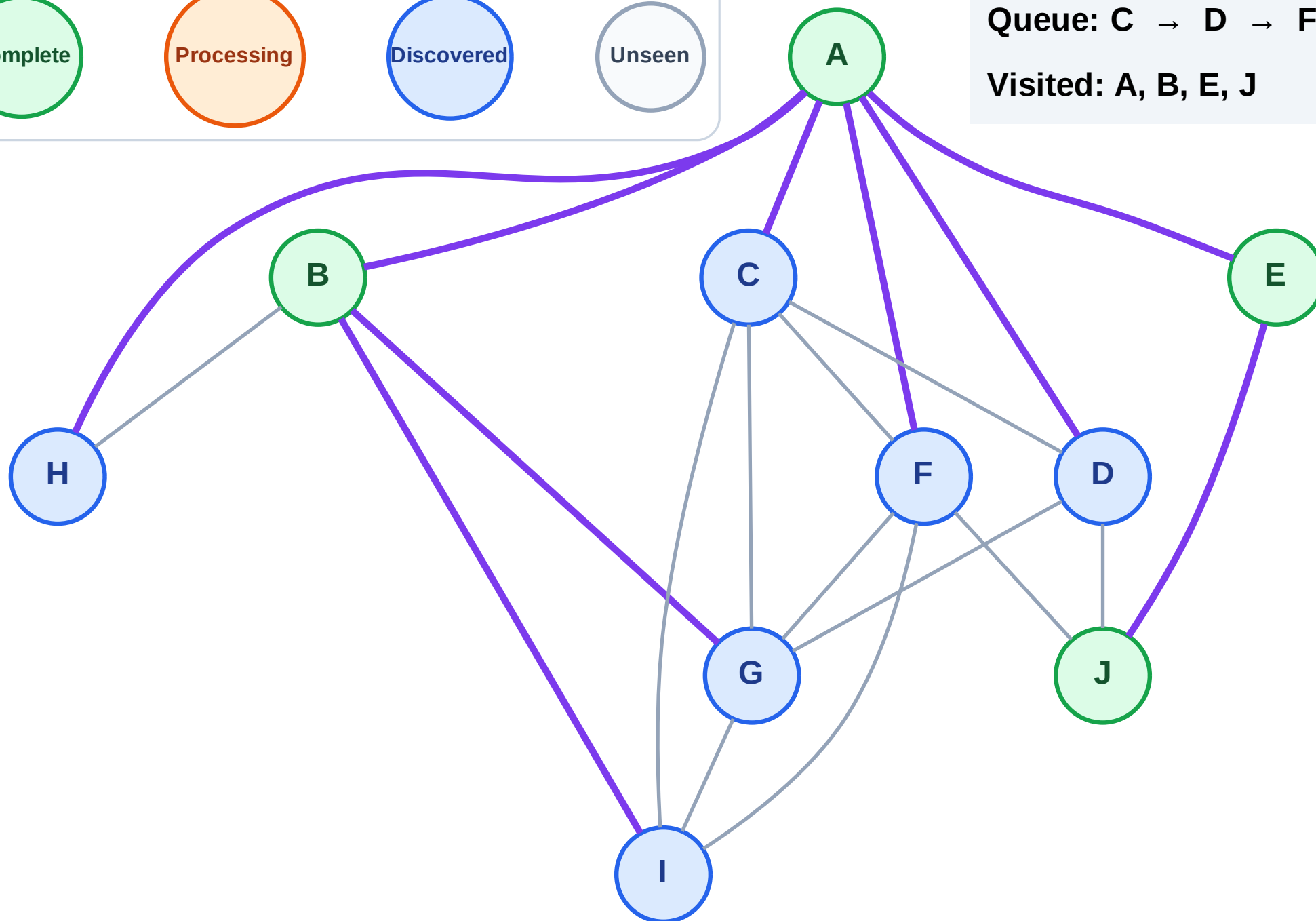
Step 9: Discover G, I from B

Legend



Queue: C → D → F → H → G → I

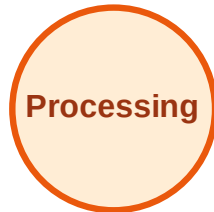
Visited: A, B, E, J



BFS Traversal

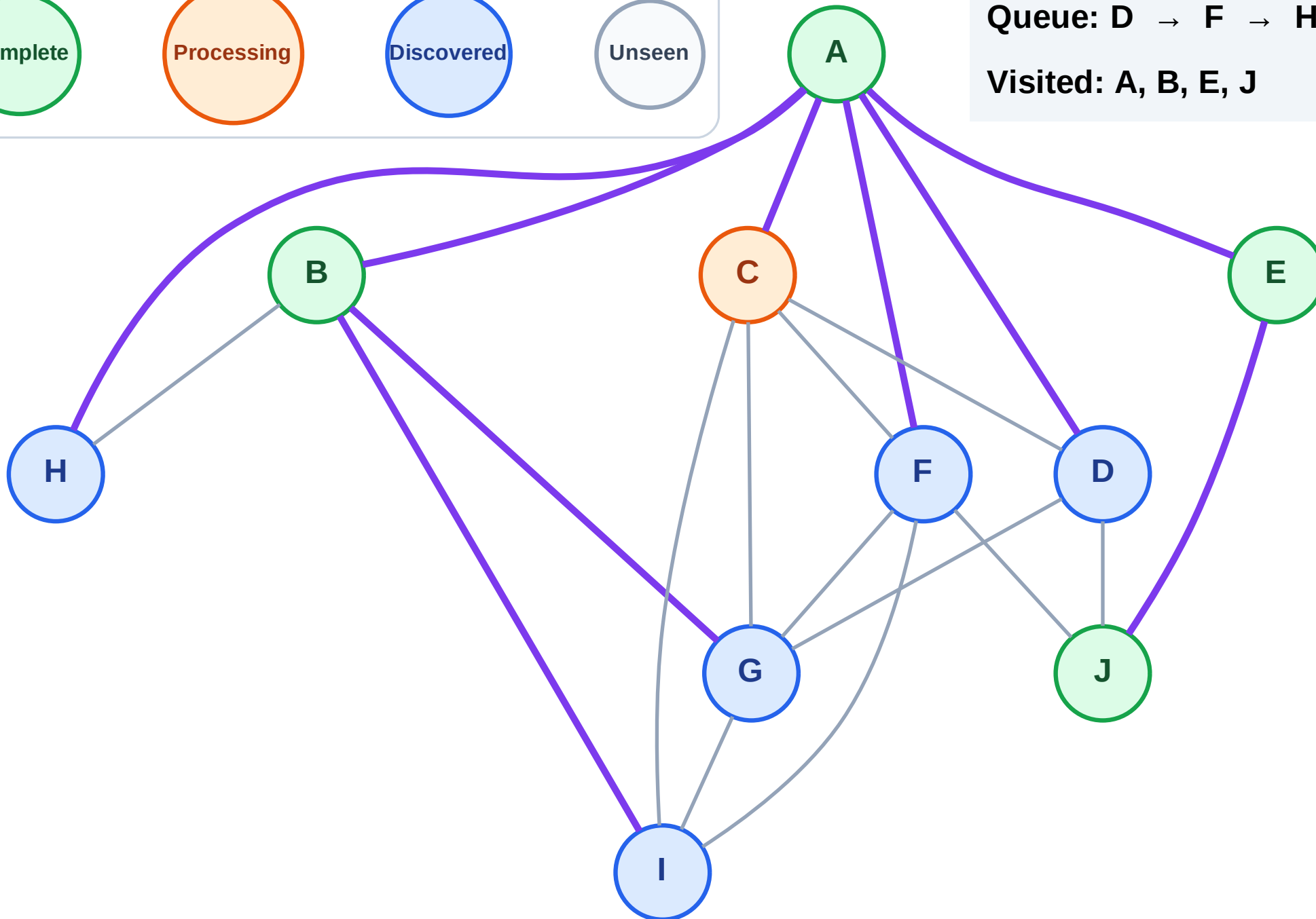
Step 10: Process node C

Legend



Queue: D → F → H → G → I

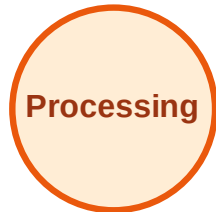
Visited: A, B, E, J



BFS Traversal

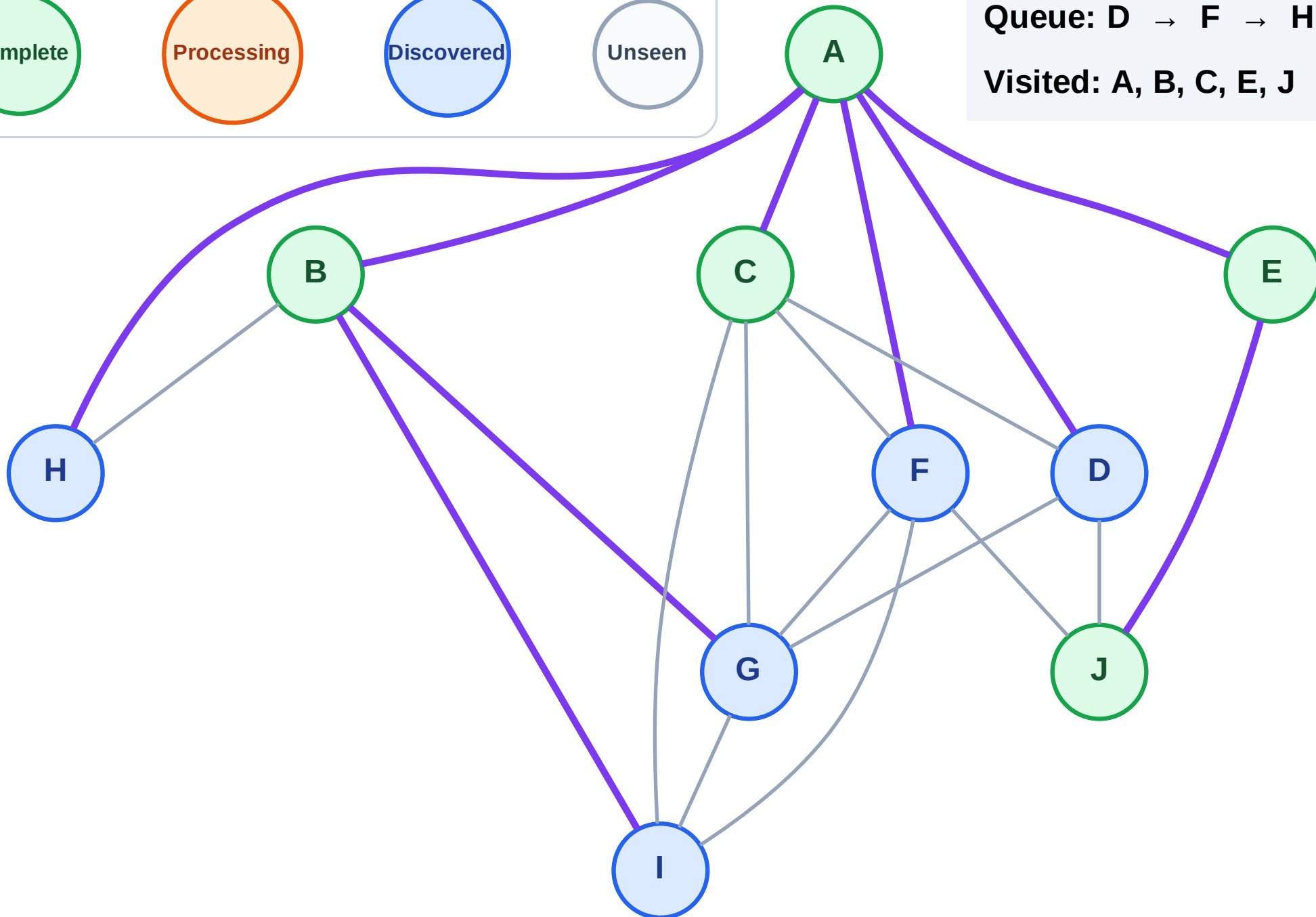
Step 11: No new neighbors from C

Legend



Queue: D → F → H → G → I

Visited: A, B, C, E, J



BFS Traversal

Step 12: Process node D

Legend

Complete

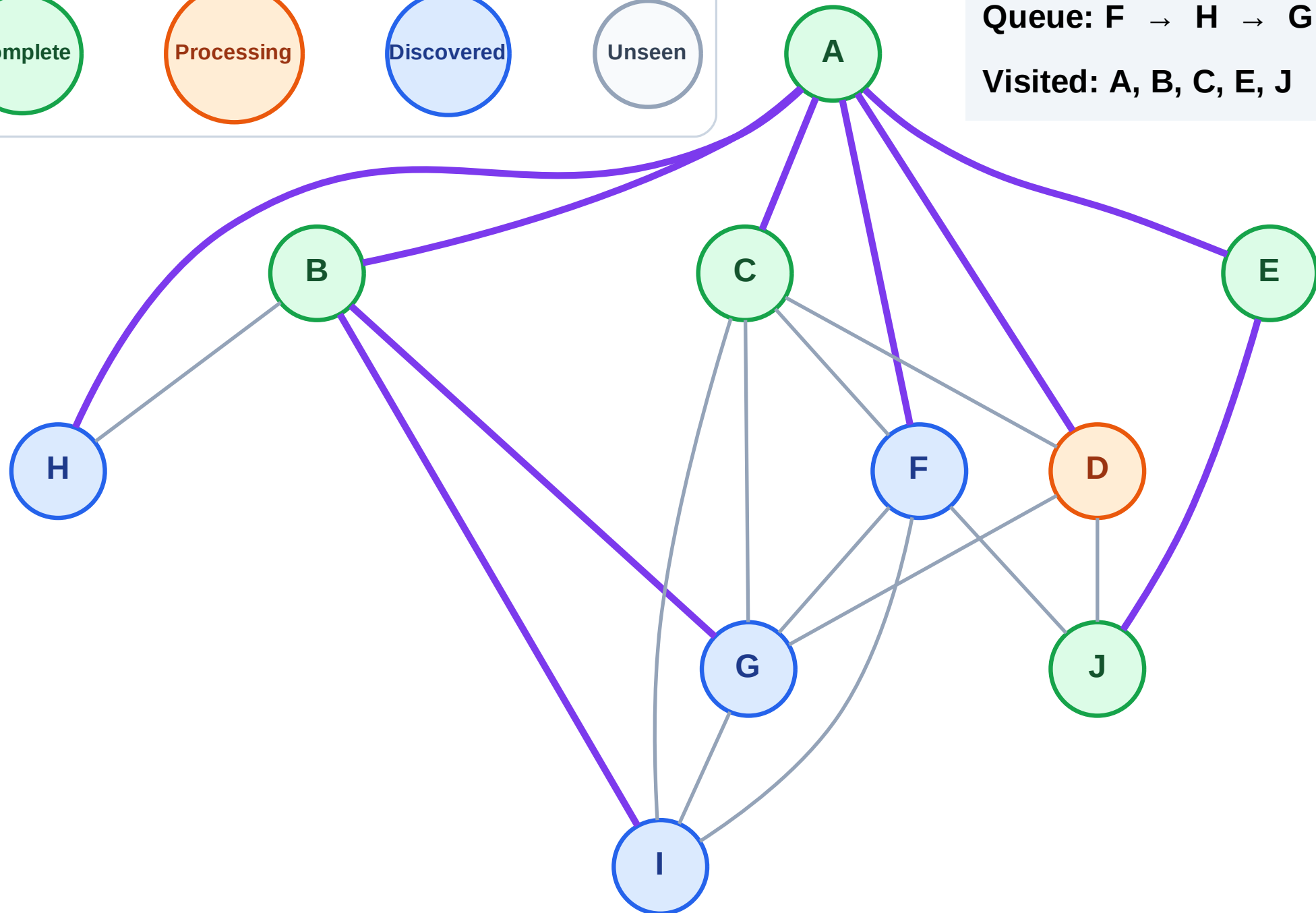
Processing

Discovered

Unseen

Queue: F → H → G → I

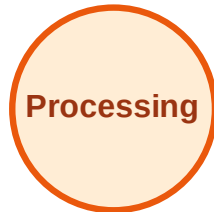
Visited: A, B, C, E, J



BFS Traversal

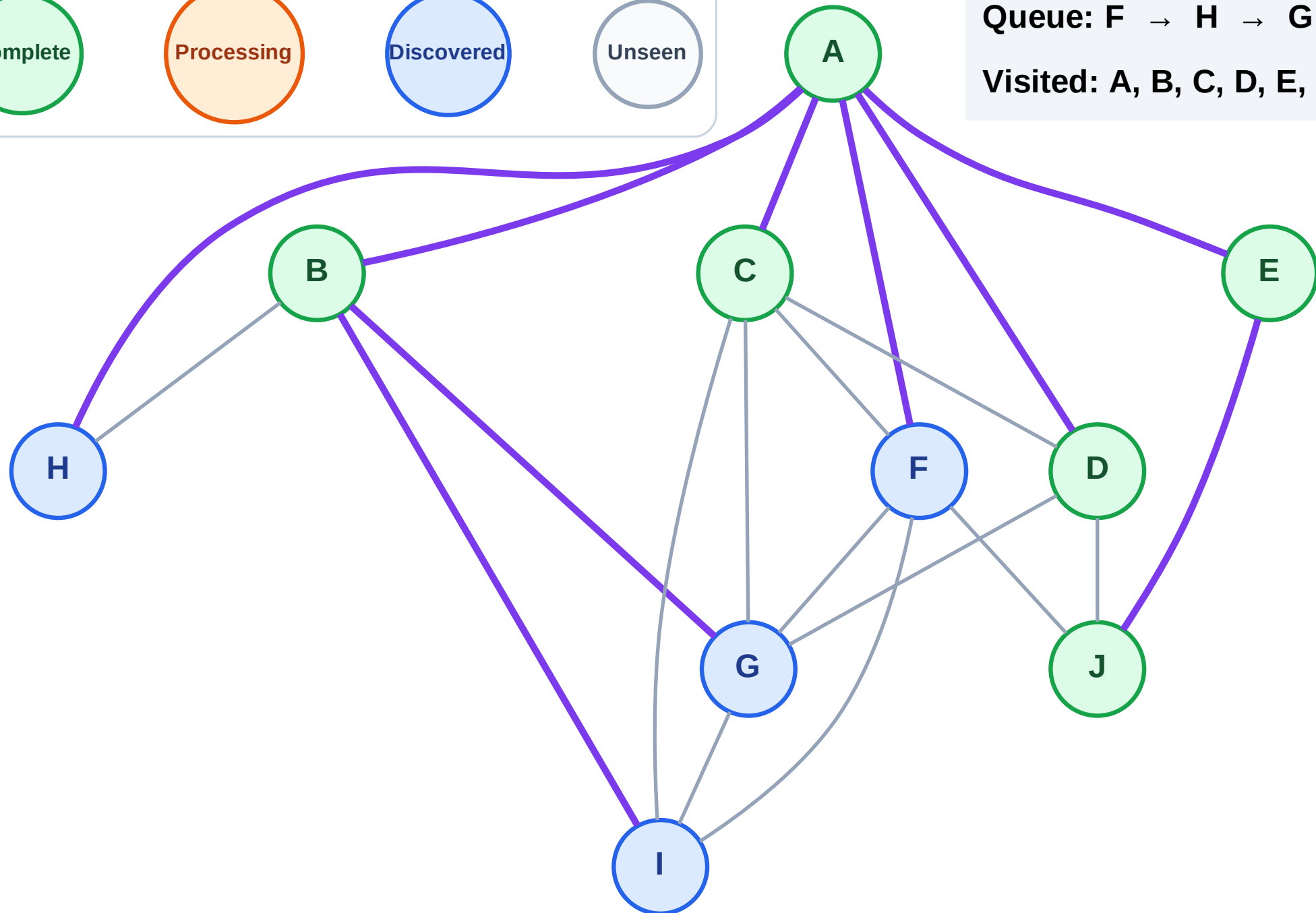
Step 13: No new neighbors from D

Legend



Queue: F → H → G → I

Visited: A, B, C, D, E, J



BFS Traversal

Step 14: Process node F

Legend

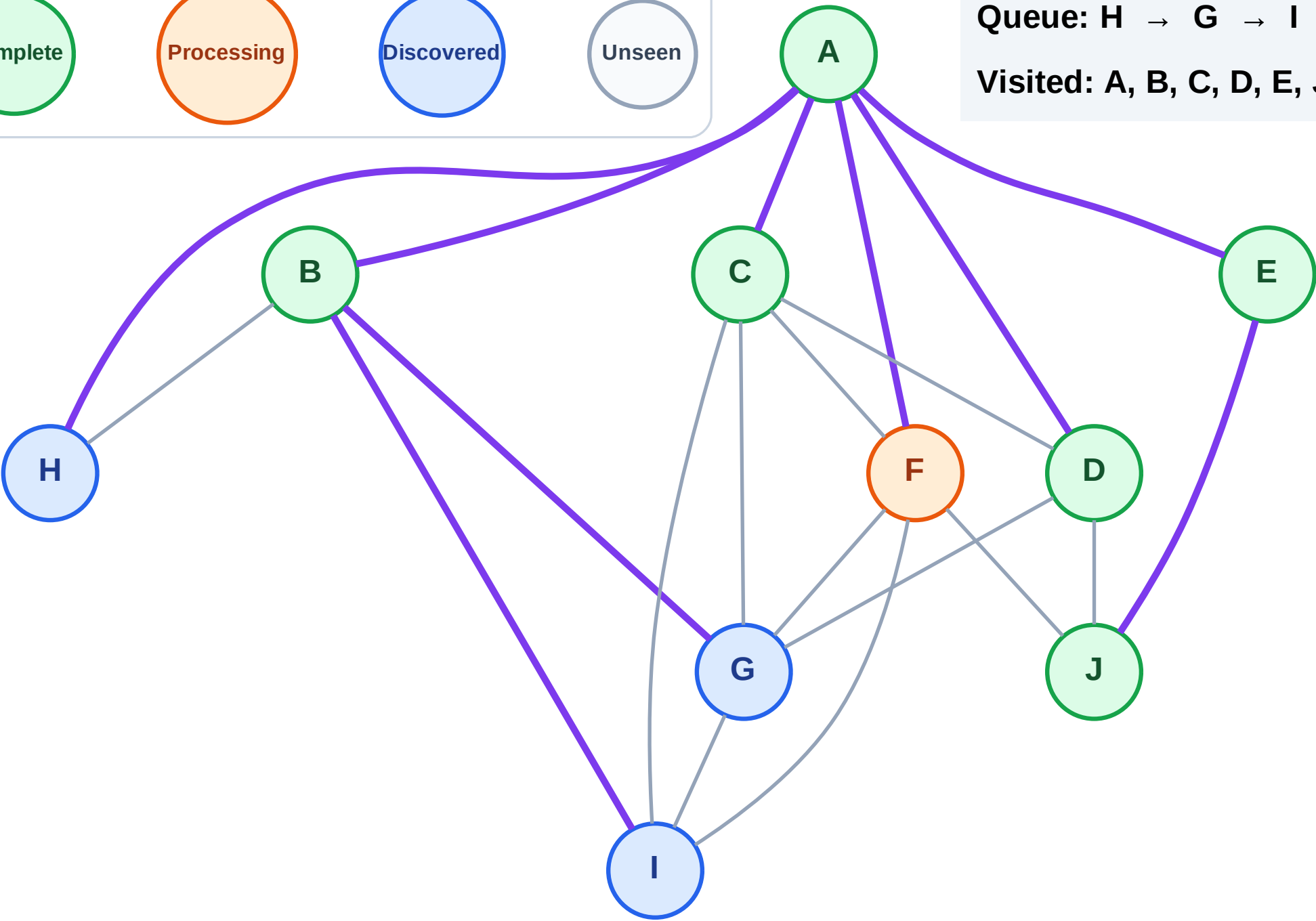
Complete

Processing

Discovered

Unseen

Queue: H → G → I
Visited: A, B, C, D, E, J



BFS Traversal

Step 15: No new neighbors from F

Legend

Complete

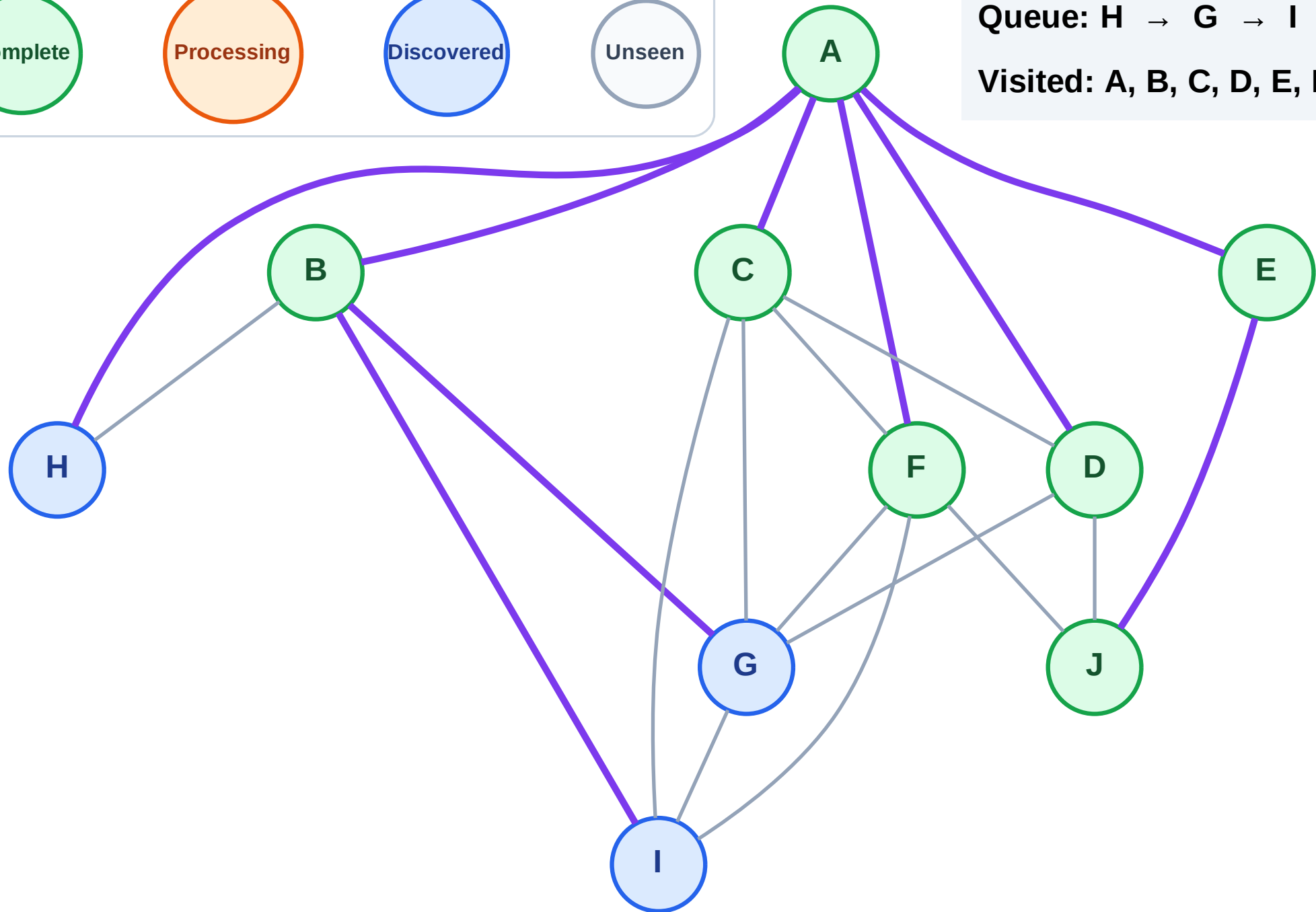
Processing

Discovered

Unseen

Queue: H → G → I

Visited: A, B, C, D, E, F, J



BFS Traversal

Step 16: Process node H

Legend

Complete

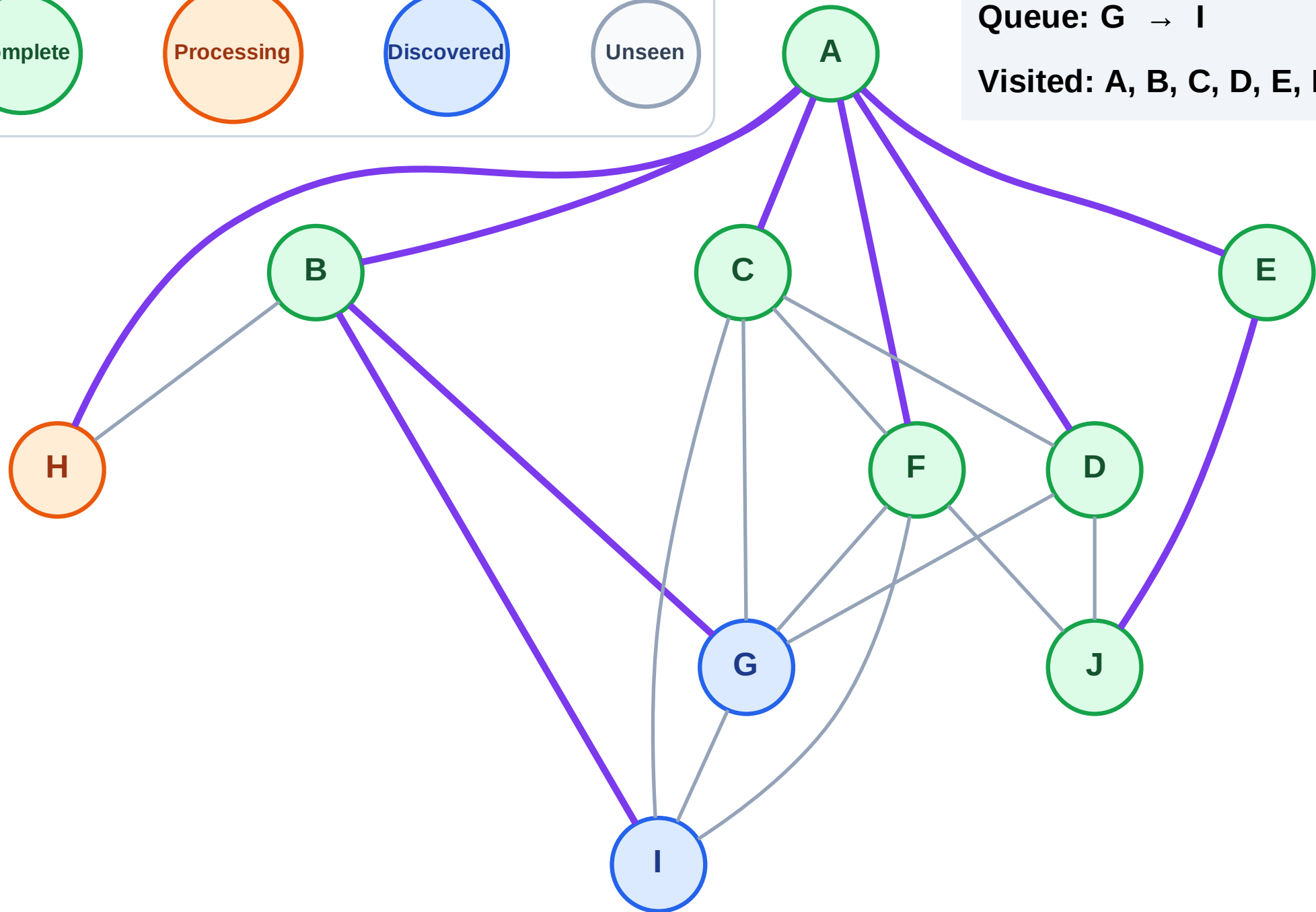
Processing

Discovered

Unseen

Queue: G → I

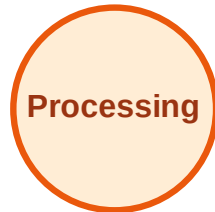
Visited: A, B, C, D, E, F, J



BFS Traversal

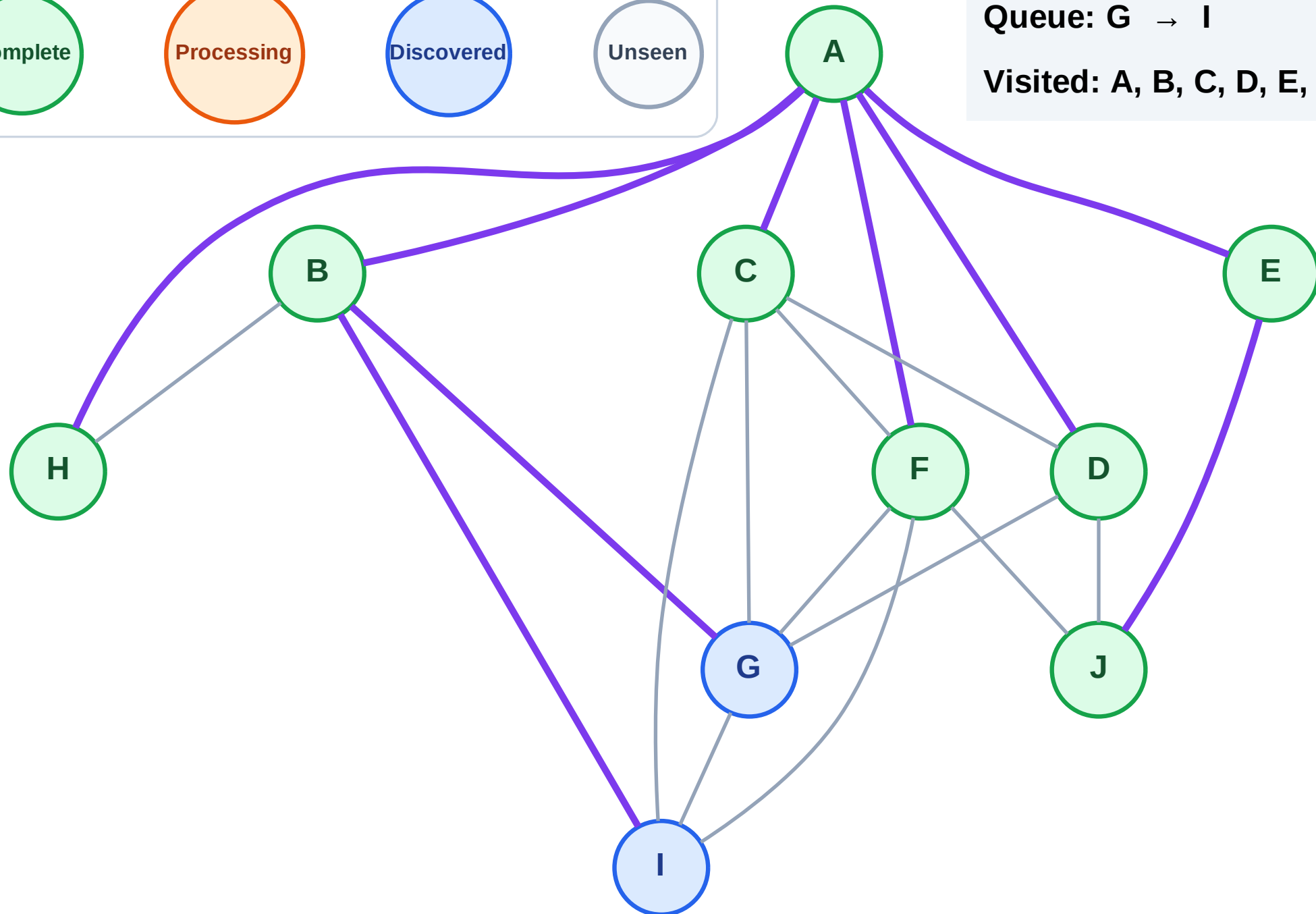
Step 17: No new neighbors from H

Legend



Queue: G → I

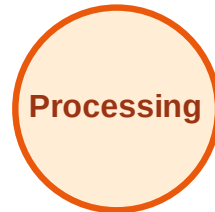
Visited: A, B, C, D, E, F, H, J



BFS Traversal

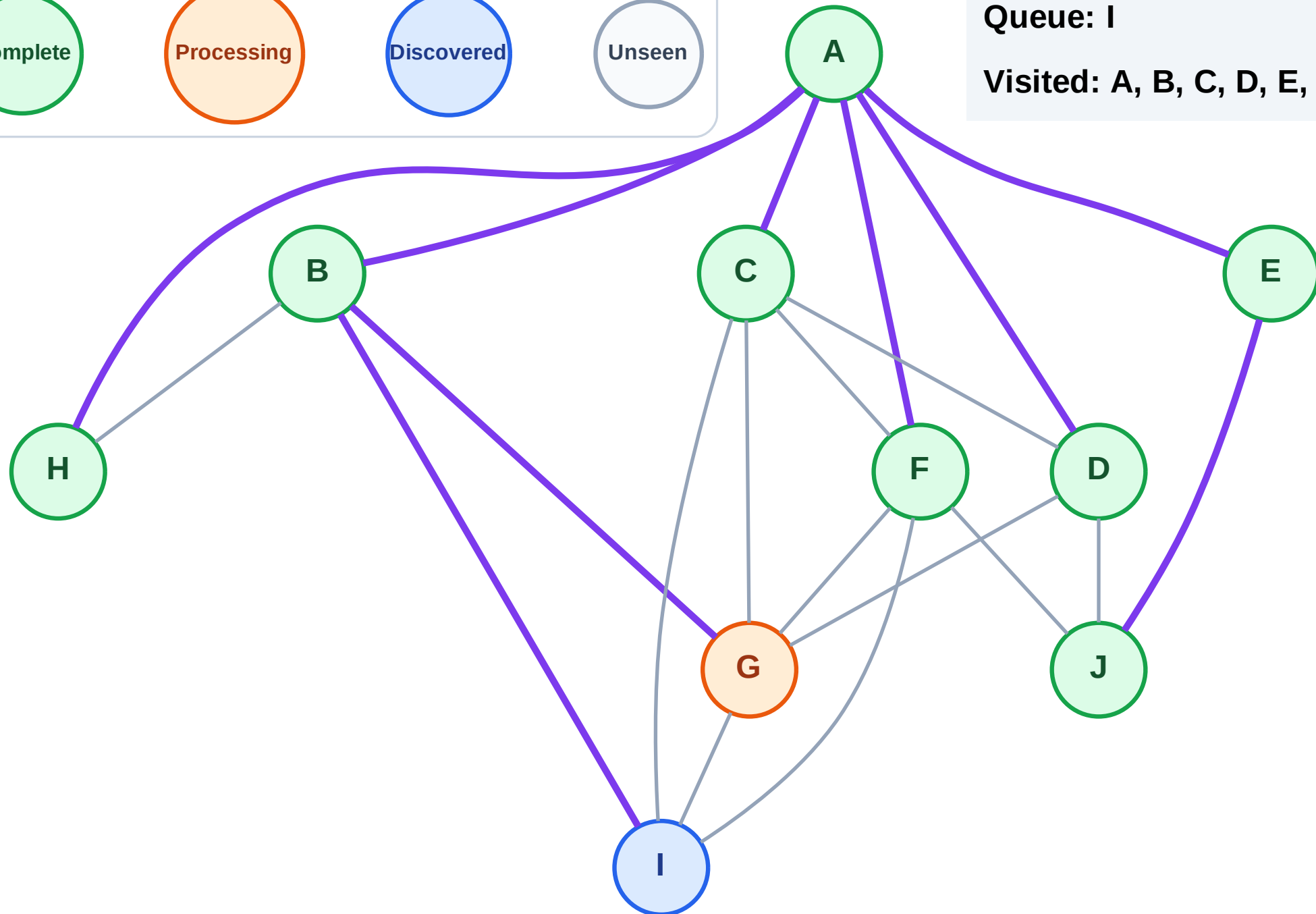
Step 18: Process node G

Legend



Queue: I

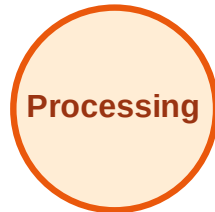
Visited: A, B, C, D, E, F, H, J



BFS Traversal

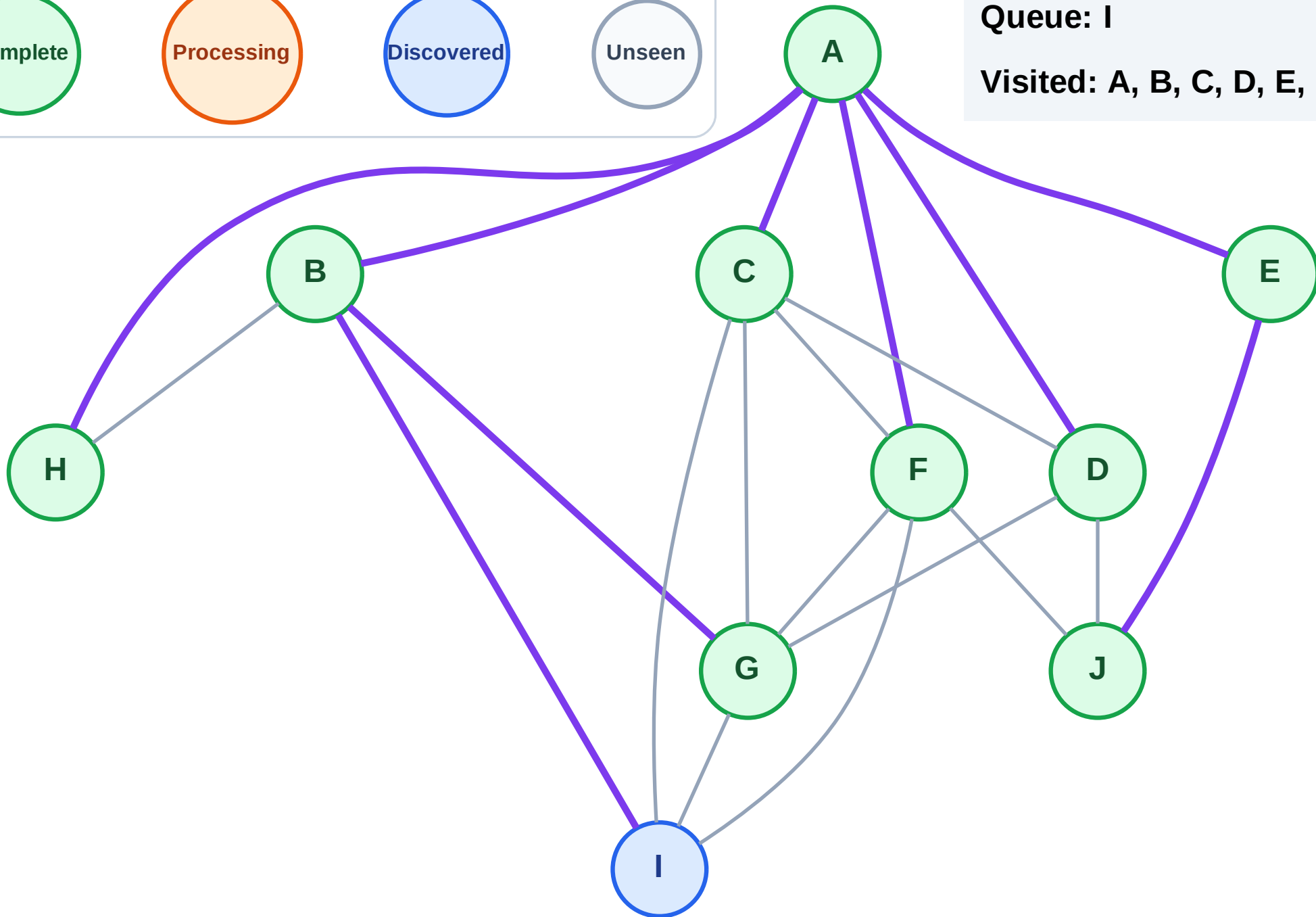
Step 19: No new neighbors from G

Legend



Queue: I

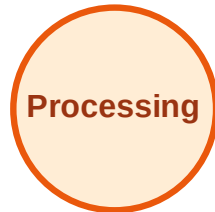
Visited: A, B, C, D, E, F, G, H, J



BFS Traversal

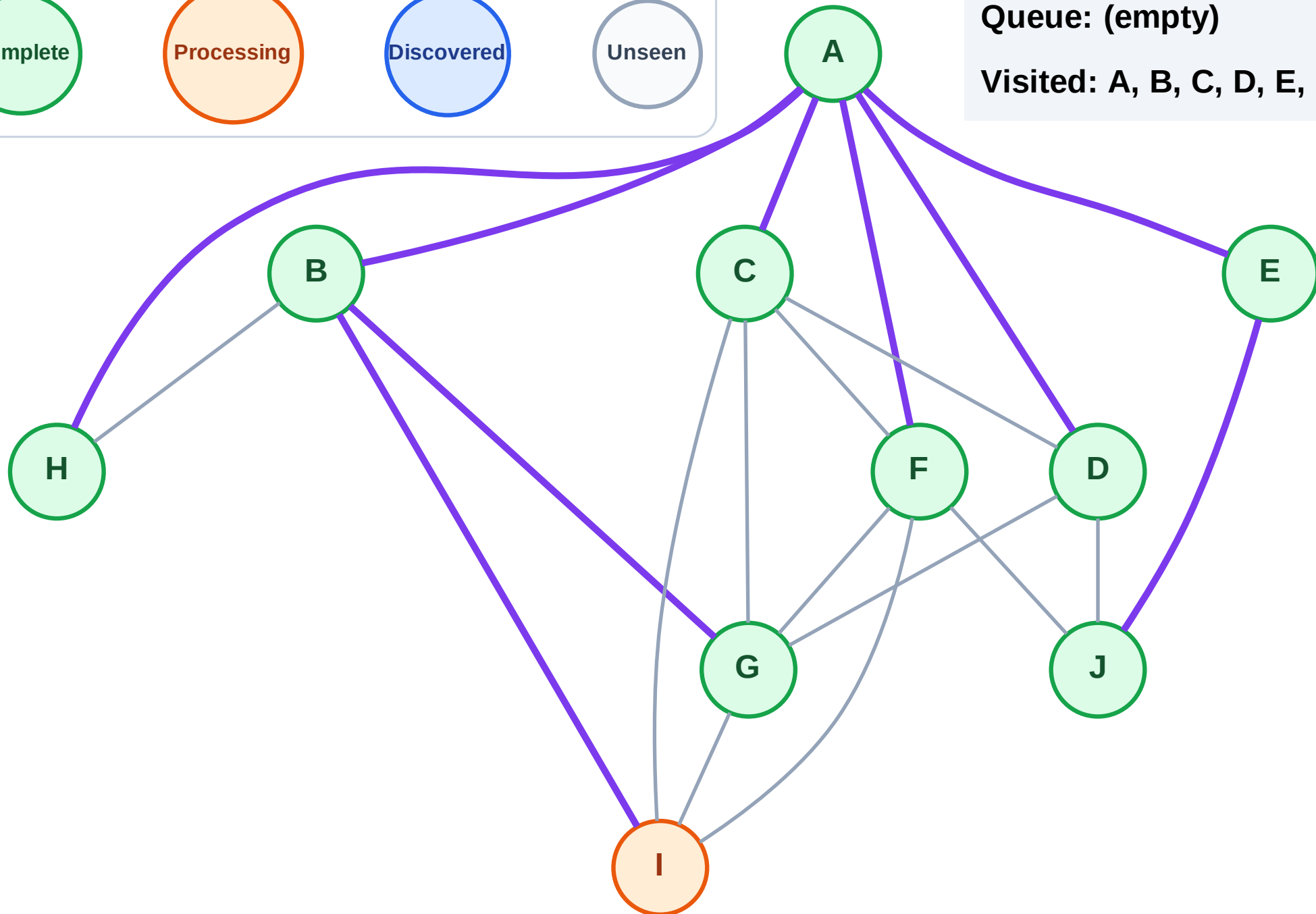
Step 20: Process node I

Legend



Queue: (empty)

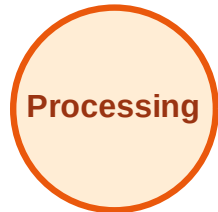
Visited: A, B, C, D, E, F, G, H, J



BFS Traversal

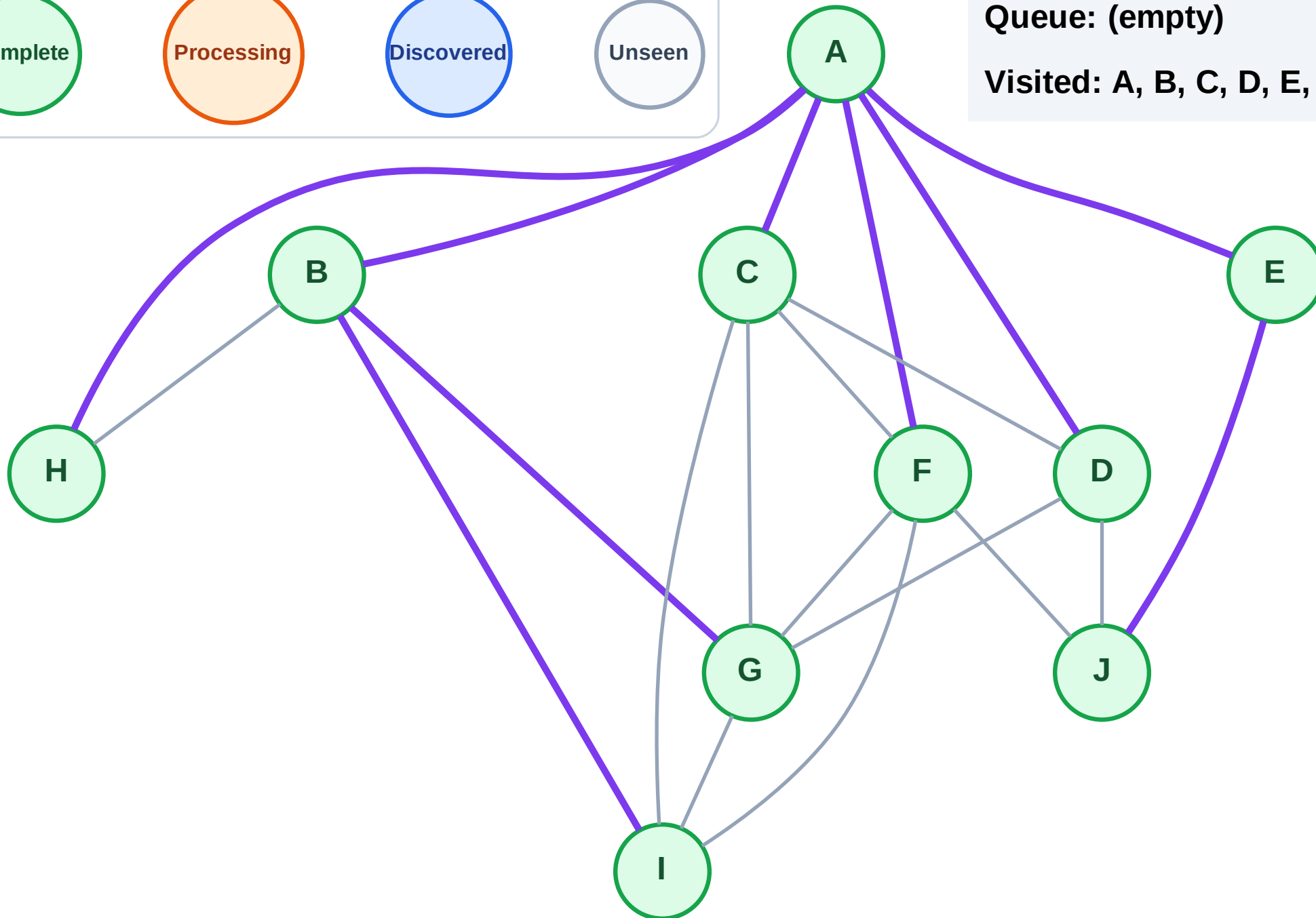
Step 21: No new neighbors from I

Legend



Queue: (empty)

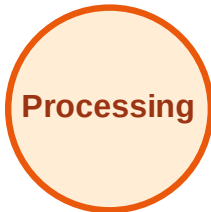
Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

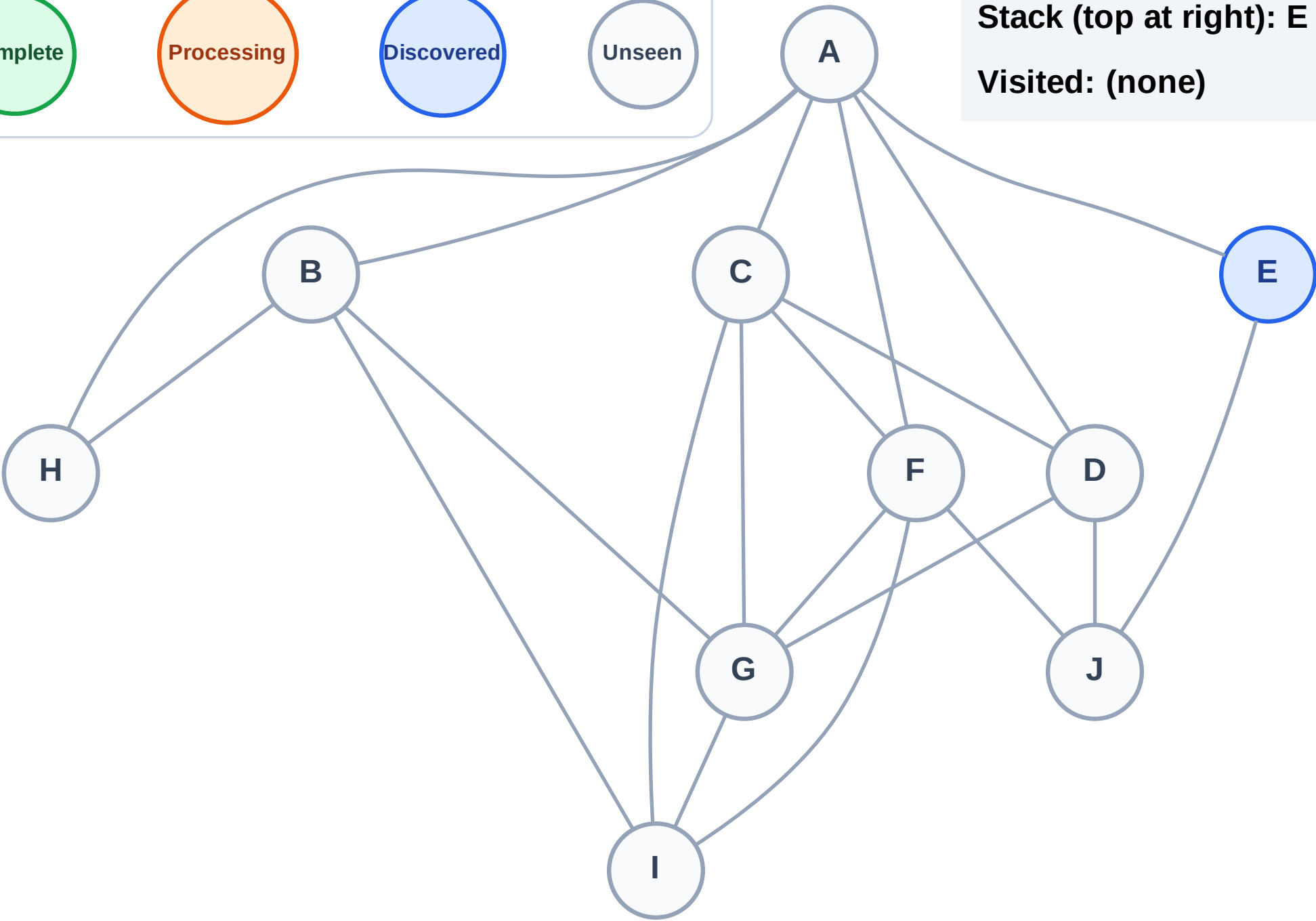
Step 1: Start at E

Legend



Stack (top at right): E

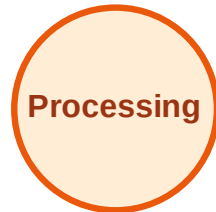
Visited: (none)



DFS Traversal

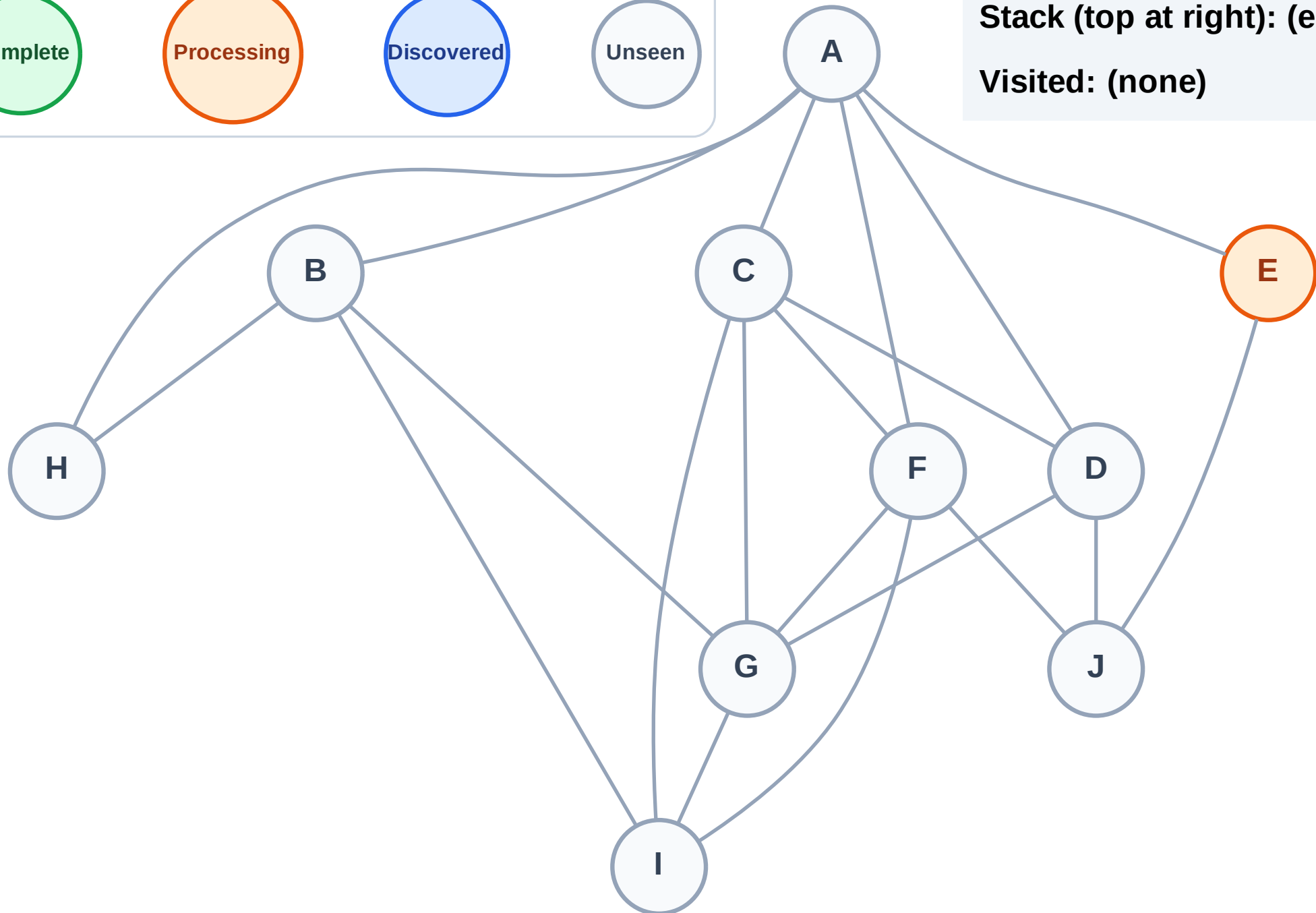
Step 2: Process node E

Legend



Stack (top at right): (empty)

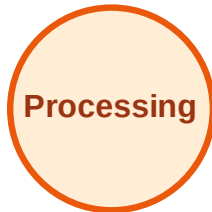
Visited: (none)



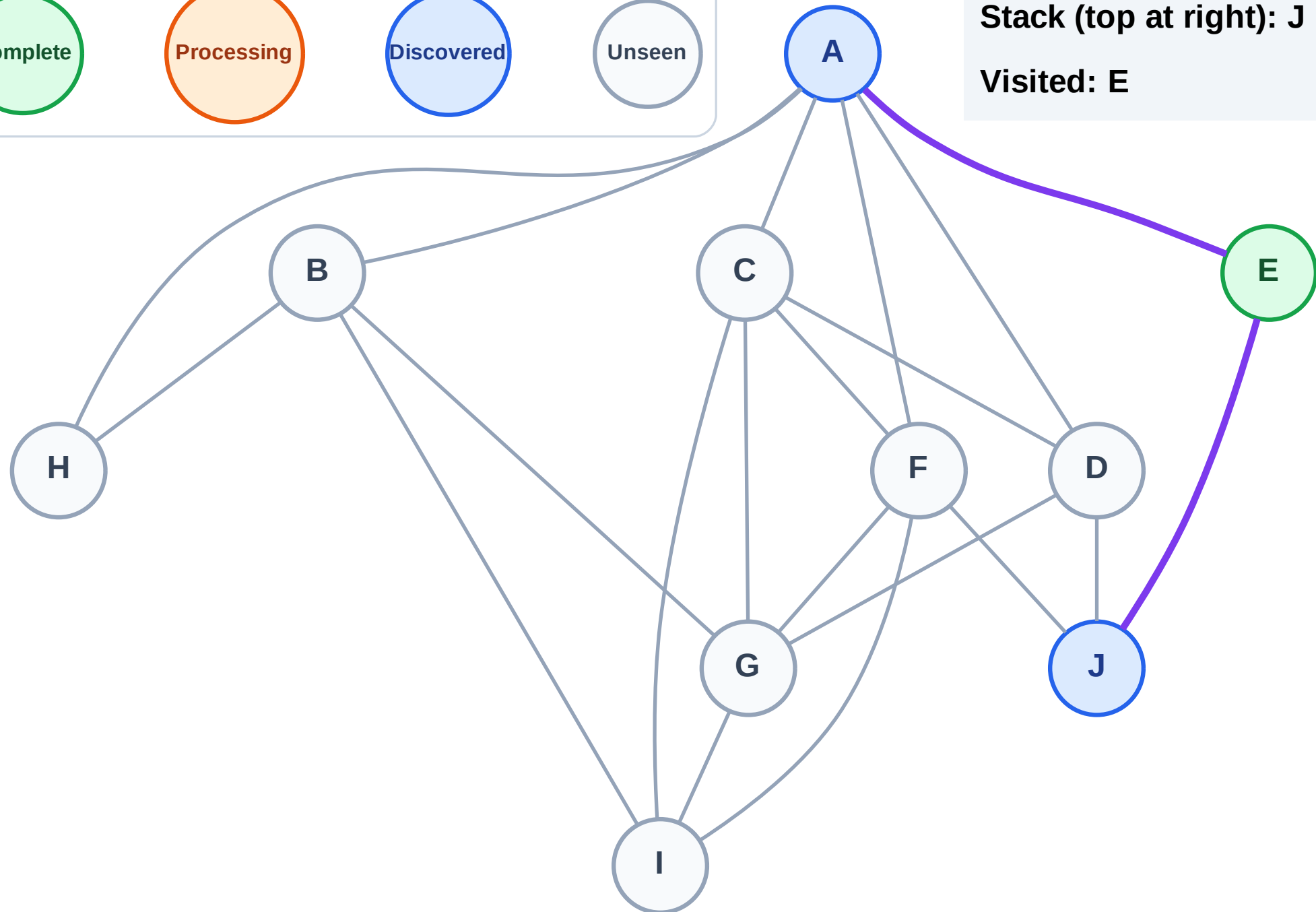
DFS Traversal

Step 3: Discover J, A from E

Legend



Stack (top at right): J | A
Visited: E



DFS Traversal

Step 4: Process node A

Legend

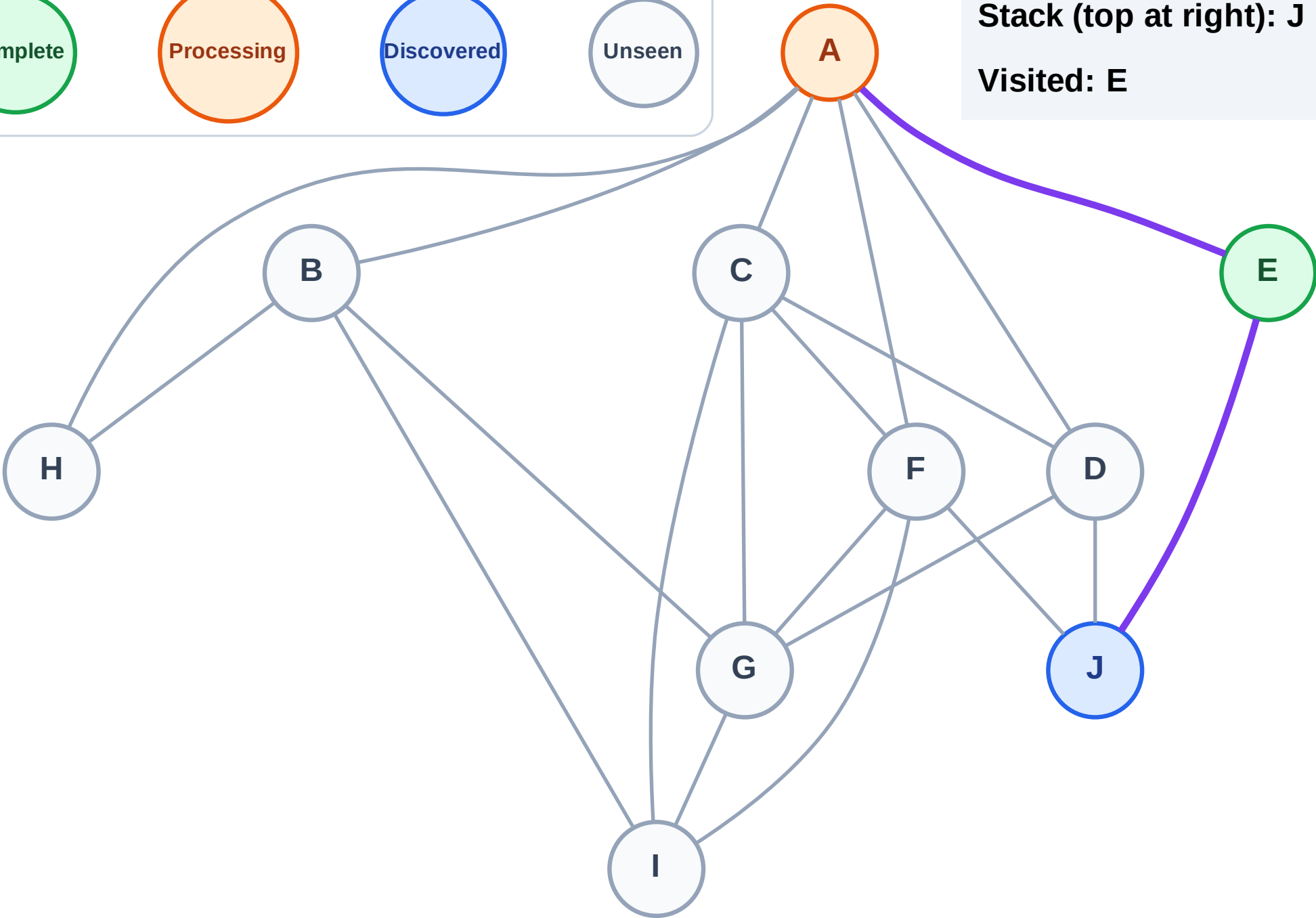
Complete

Processing

Discovered

Unseen

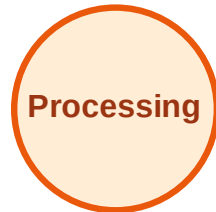
Stack (top at right): J
Visited: E



DFS Traversal

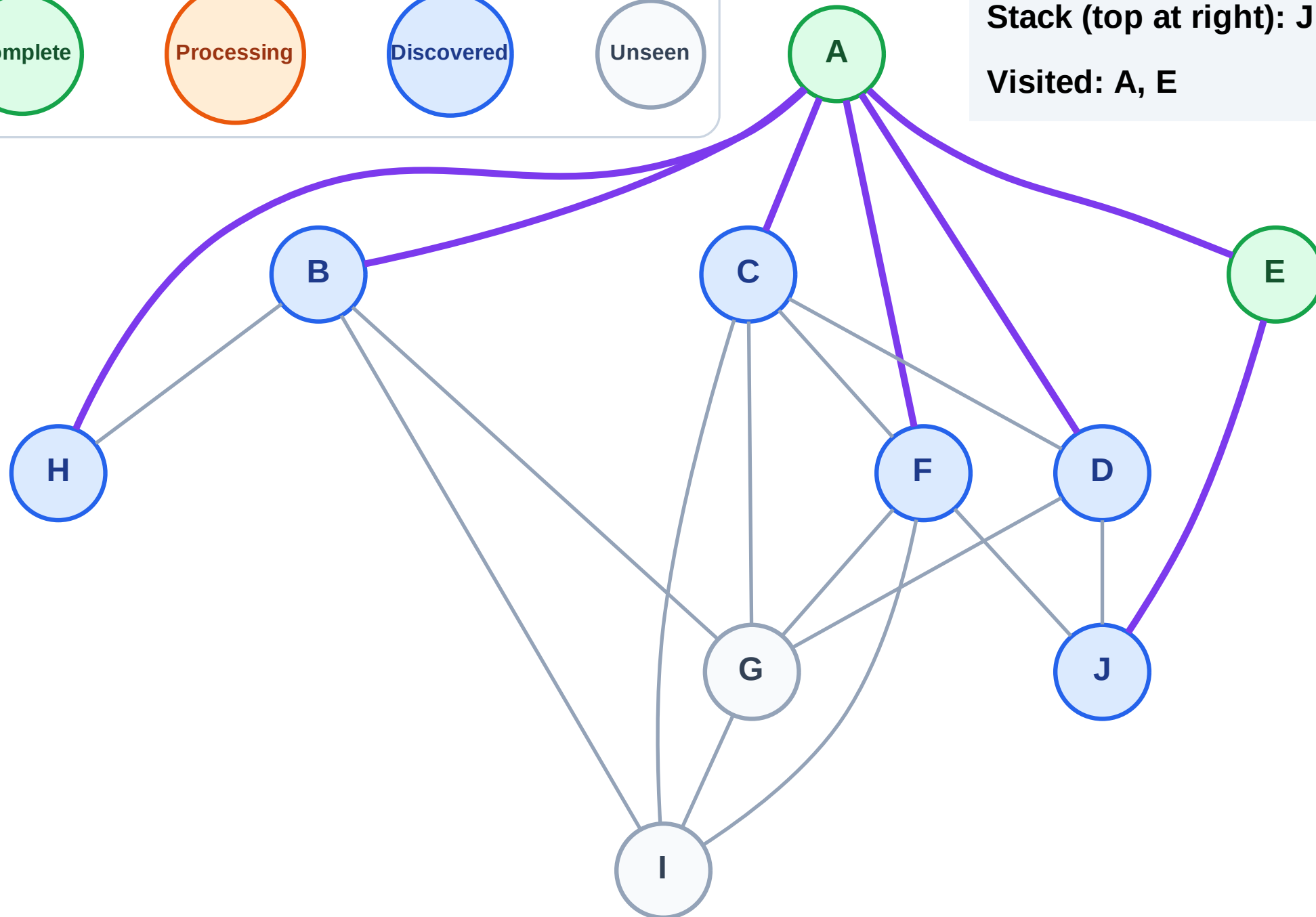
Step 5: Discover H, F, D, C, B from A

Legend



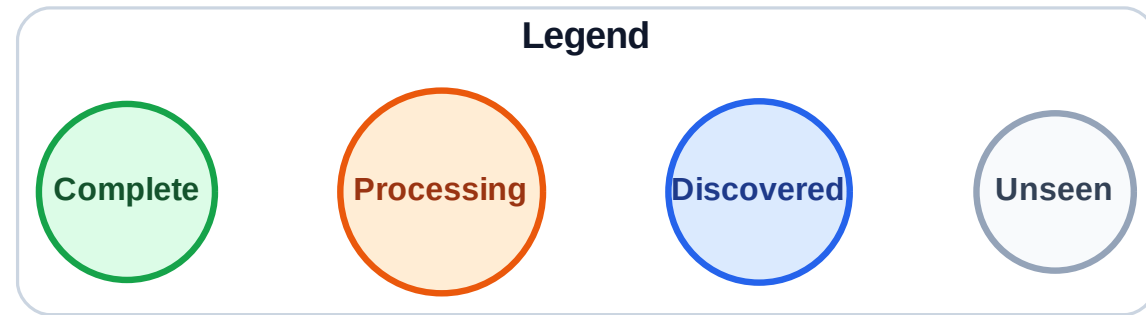
Stack (top at right): J | H | F | D | C | B

Visited: A, E

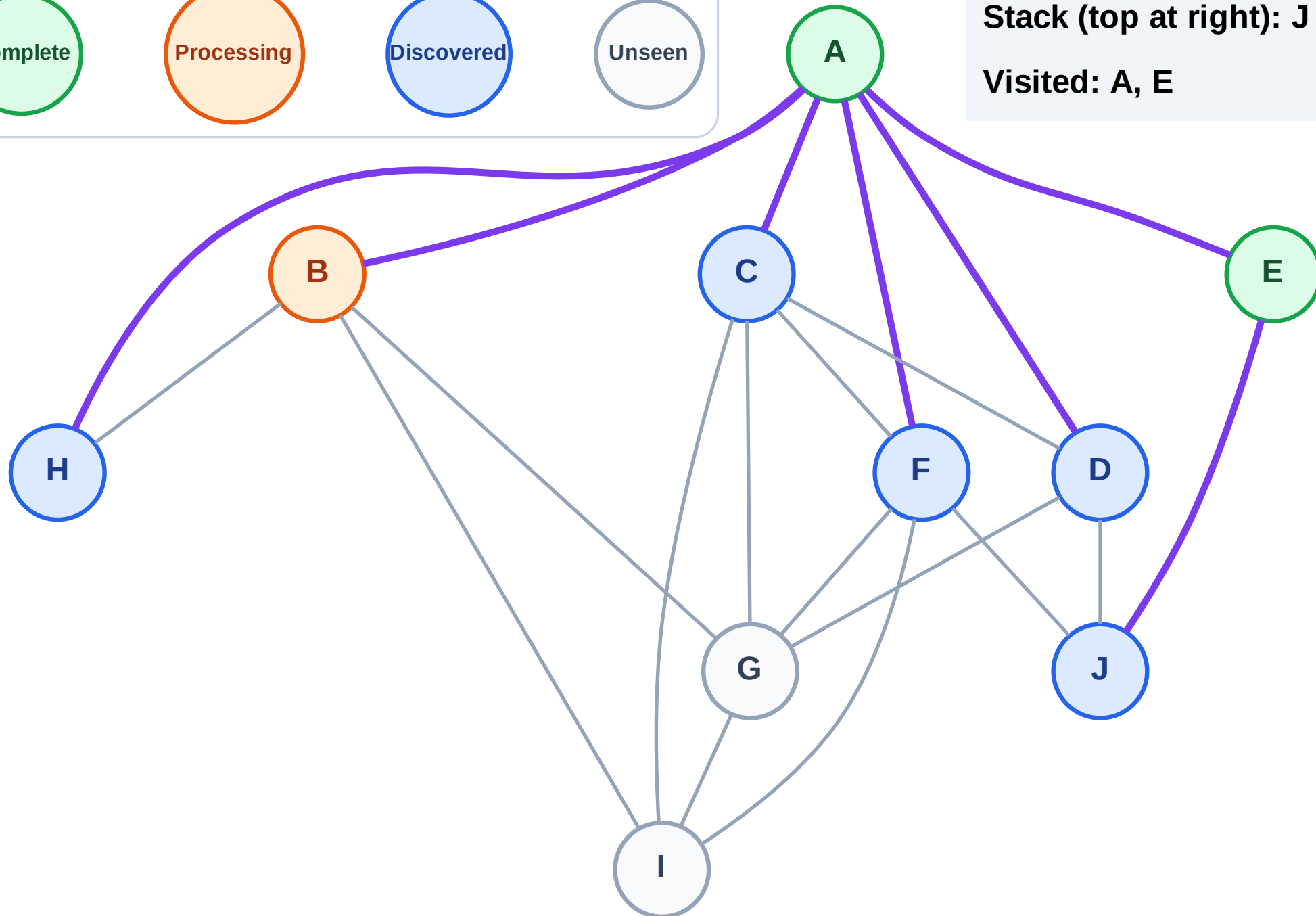


DFS Traversal

Step 6: Process node B



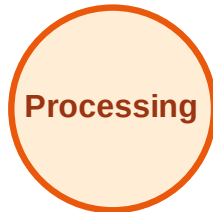
Stack (top at right): J | H | F | D | C
Visited: A, E



DFS Traversal

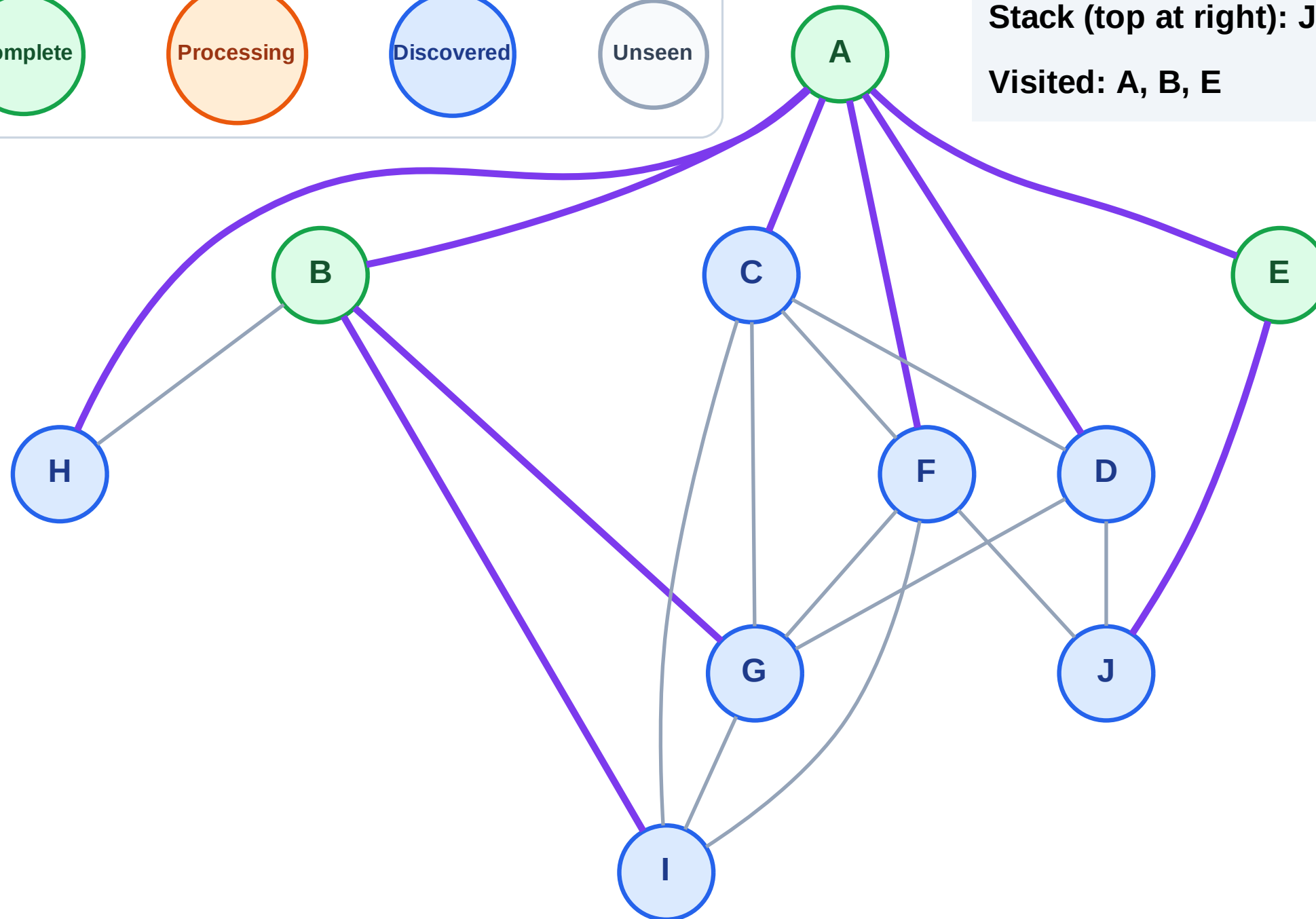
Step 7: Discover I, G from B

Legend



Stack (top at right): J | H | F | D | C | I | G

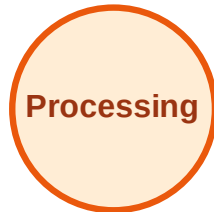
Visited: A, B, E



DFS Traversal

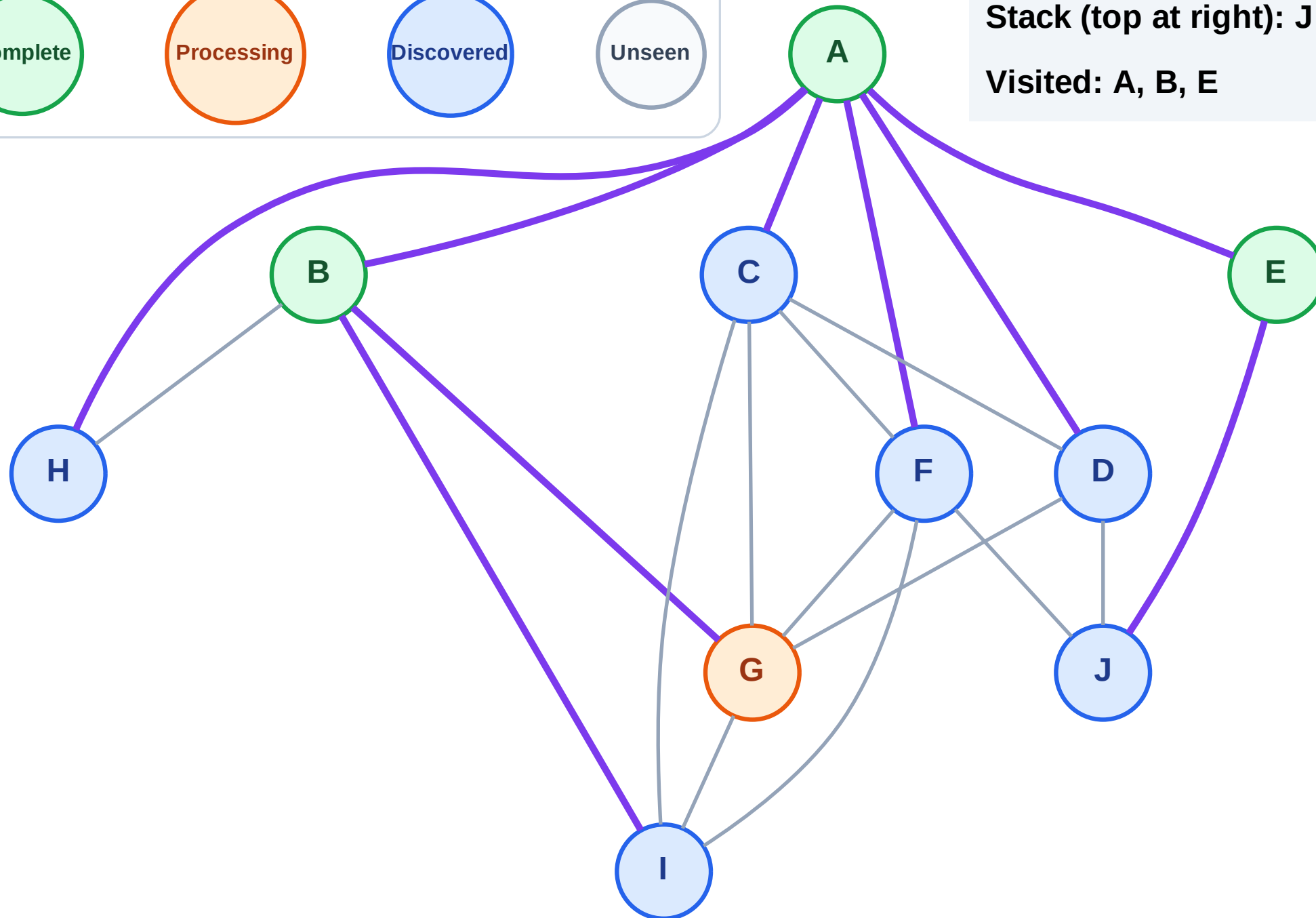
Step 8: Process node G

Legend



Stack (top at right): J | H | F | D | C | I

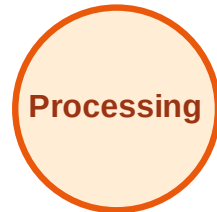
Visited: A, B, E



DFS Traversal

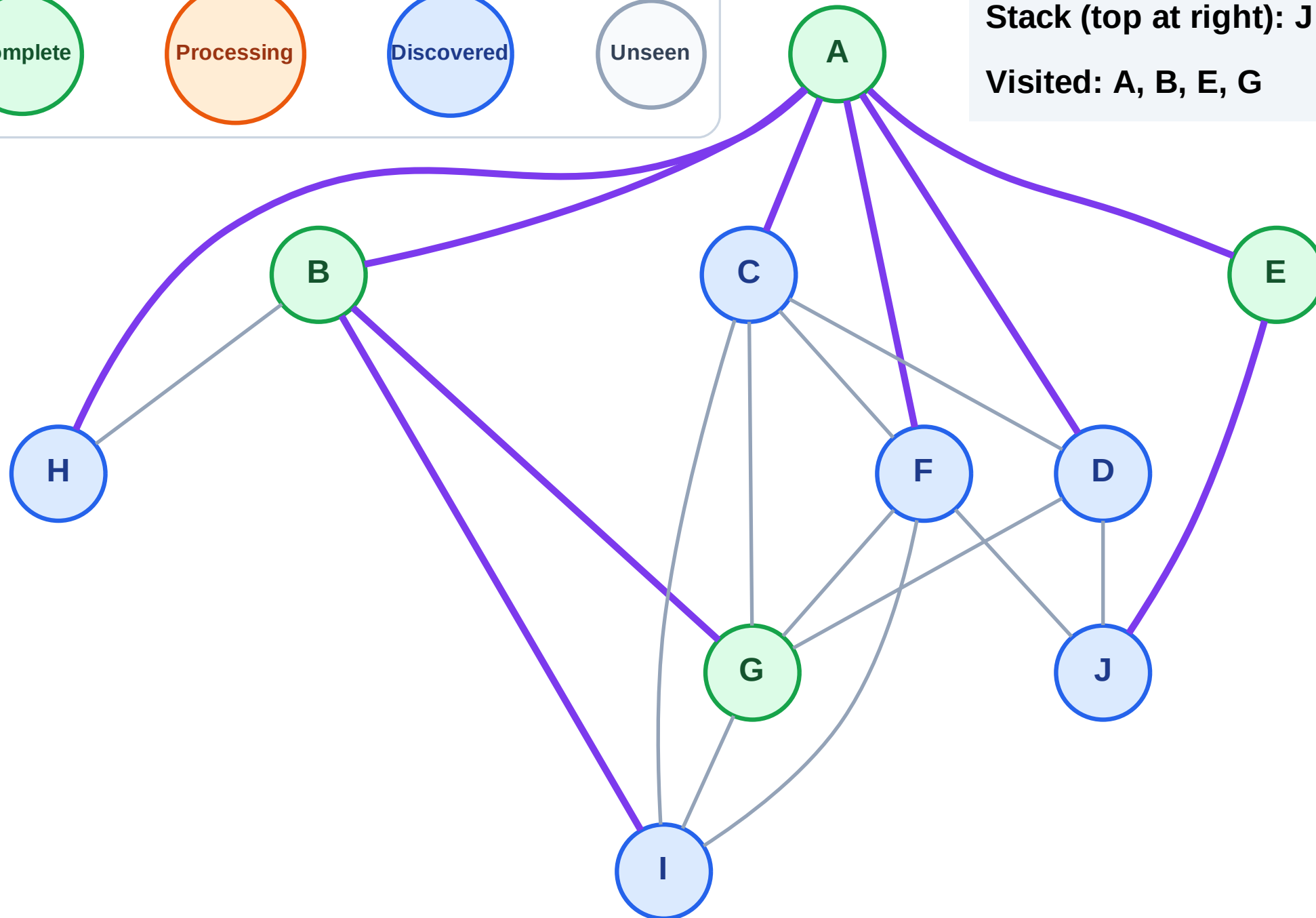
Step 9: No new neighbors from G

Legend



Stack (top at right): J | H | F | D | C | I

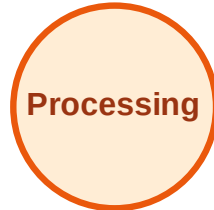
Visited: A, B, E, G



DFS Traversal

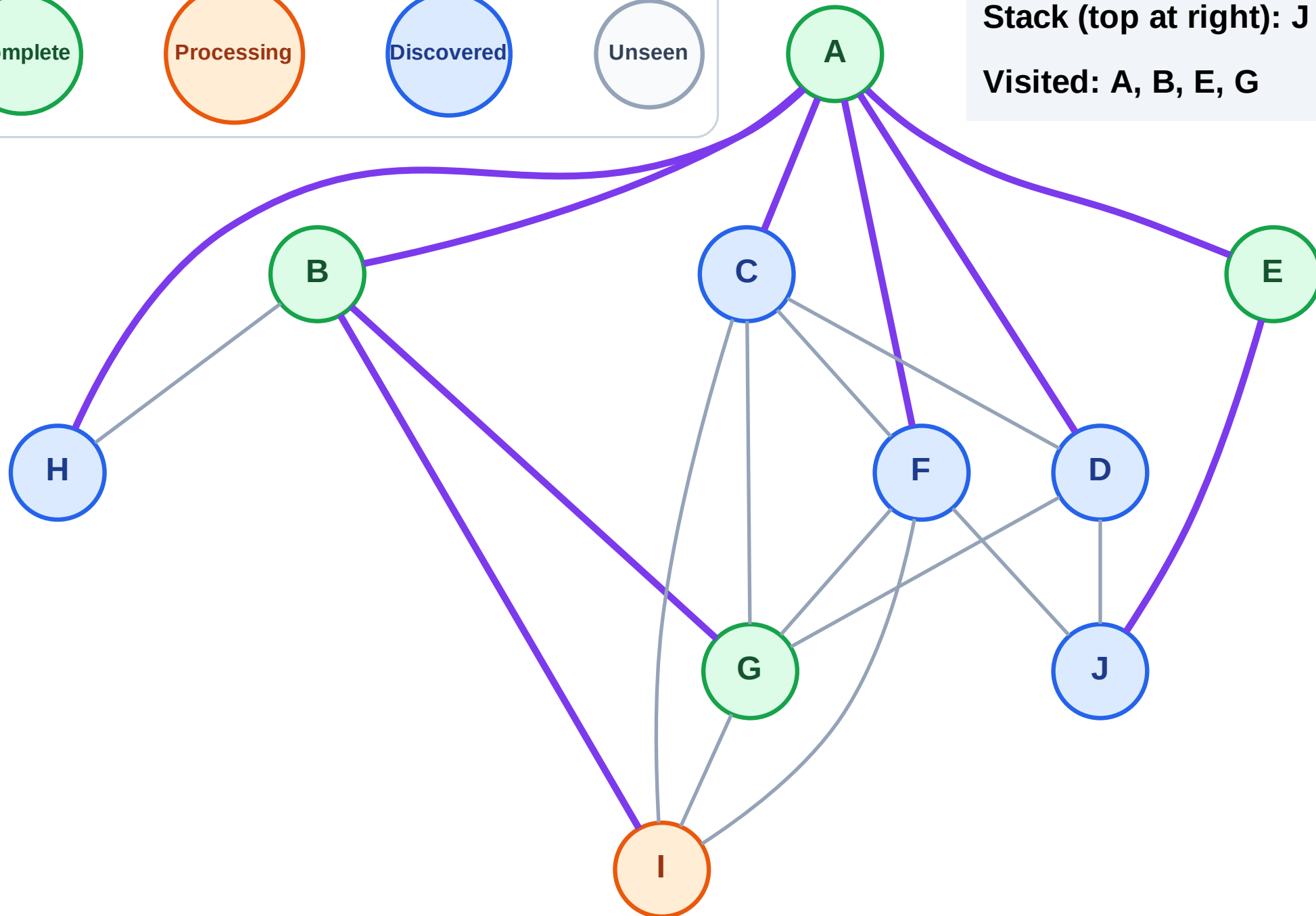
Step 10: Process node I

Legend



Stack (top at right): J | H | F | D | C

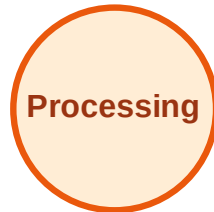
Visited: A, B, E, G



DFS Traversal

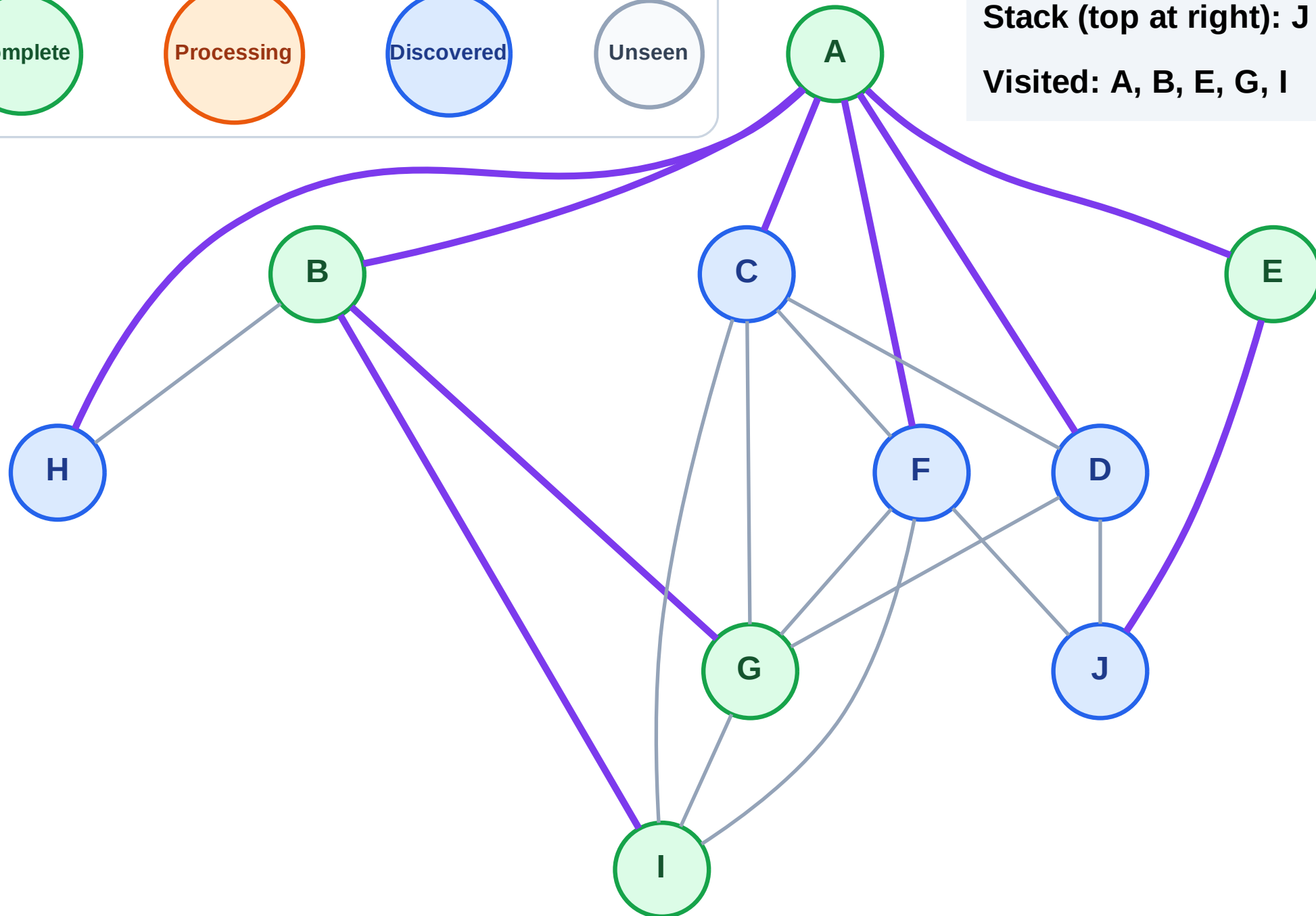
Step 11: No new neighbors from I

Legend



Stack (top at right): J | H | F | D | C

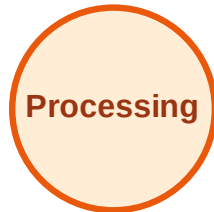
Visited: A, B, E, G, I



DFS Traversal

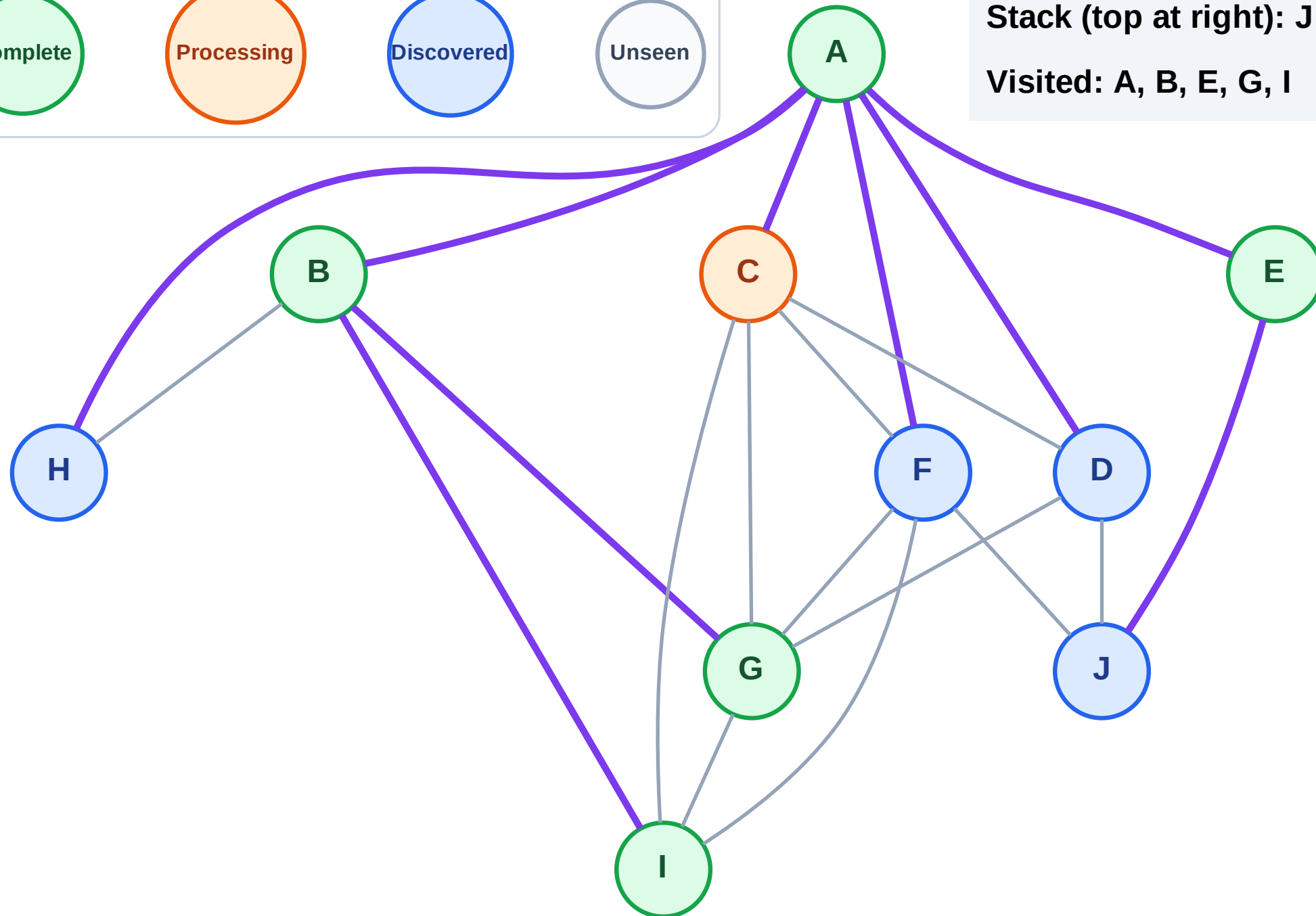
Step 12: Process node C

Legend



Stack (top at right): J | H | F | D

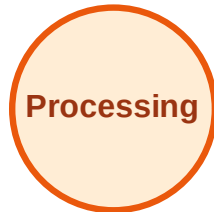
Visited: A, B, E, G, I



DFS Traversal

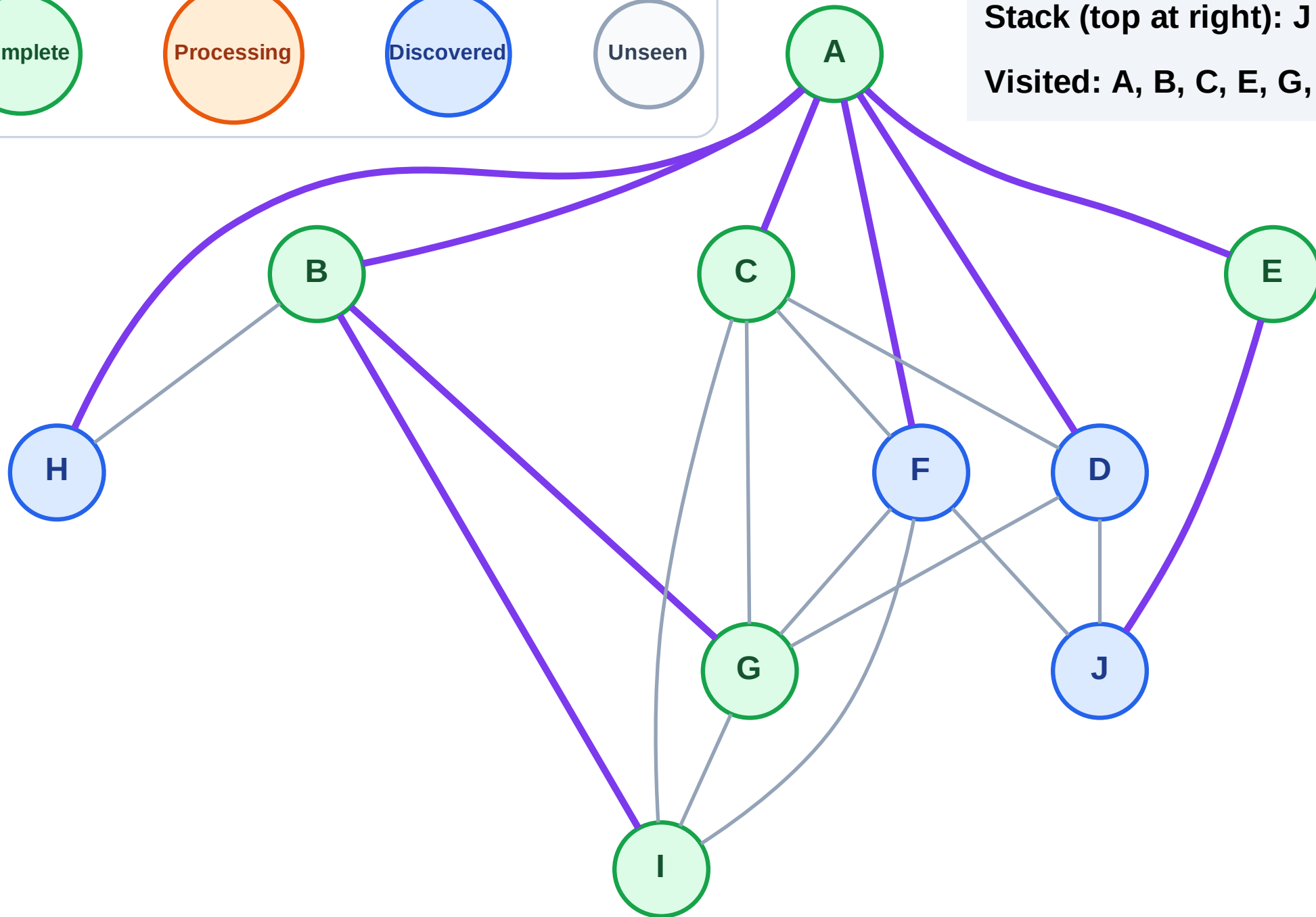
Step 13: No new neighbors from C

Legend



Stack (top at right): J | H | F | D

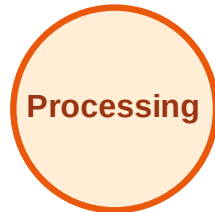
Visited: A, B, C, E, G, I



DFS Traversal

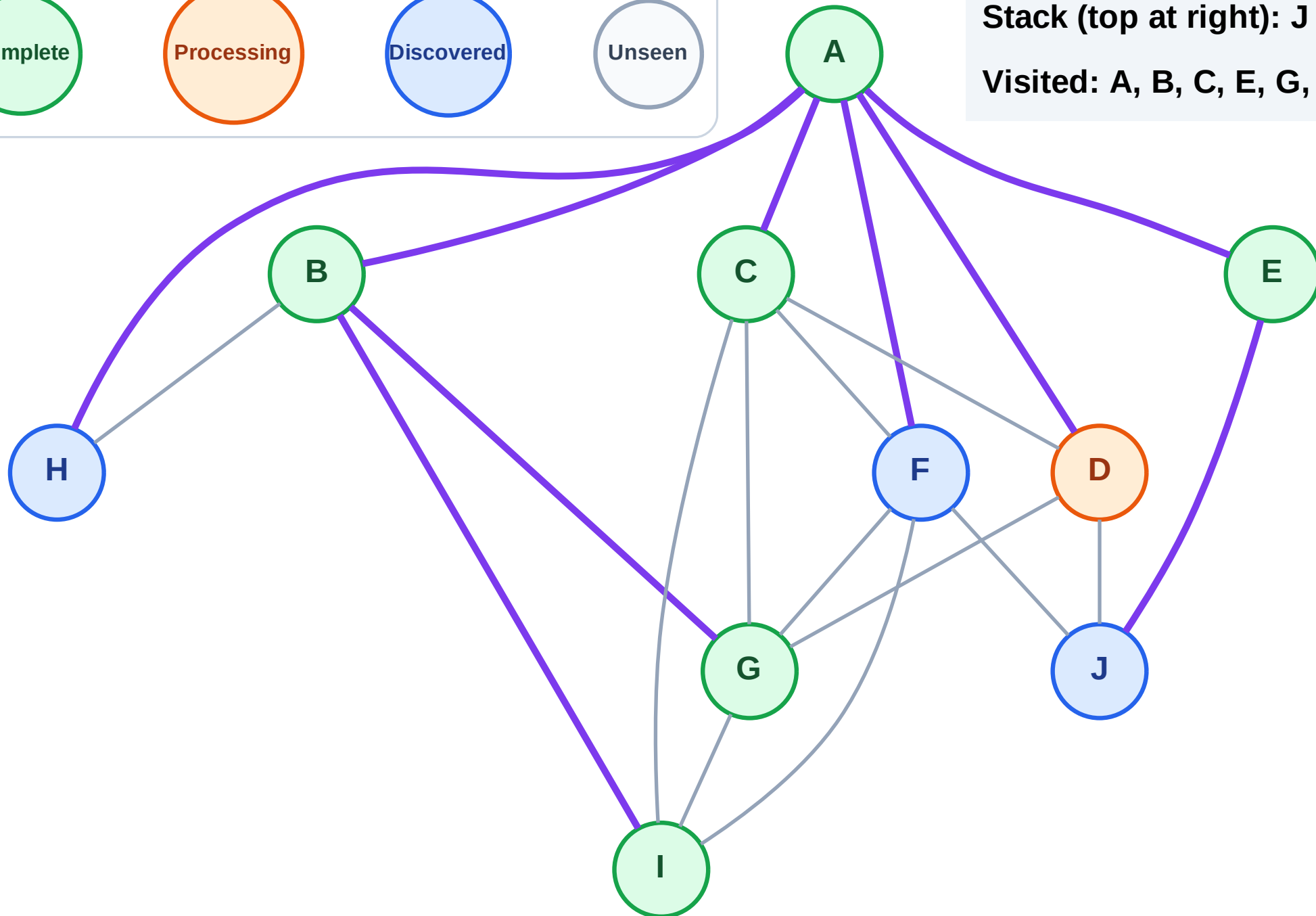
Step 14: Process node D

Legend



Stack (top at right): J | H | F

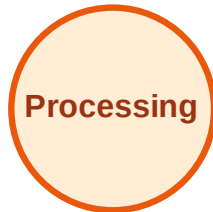
Visited: A, B, C, E, G, I



DFS Traversal

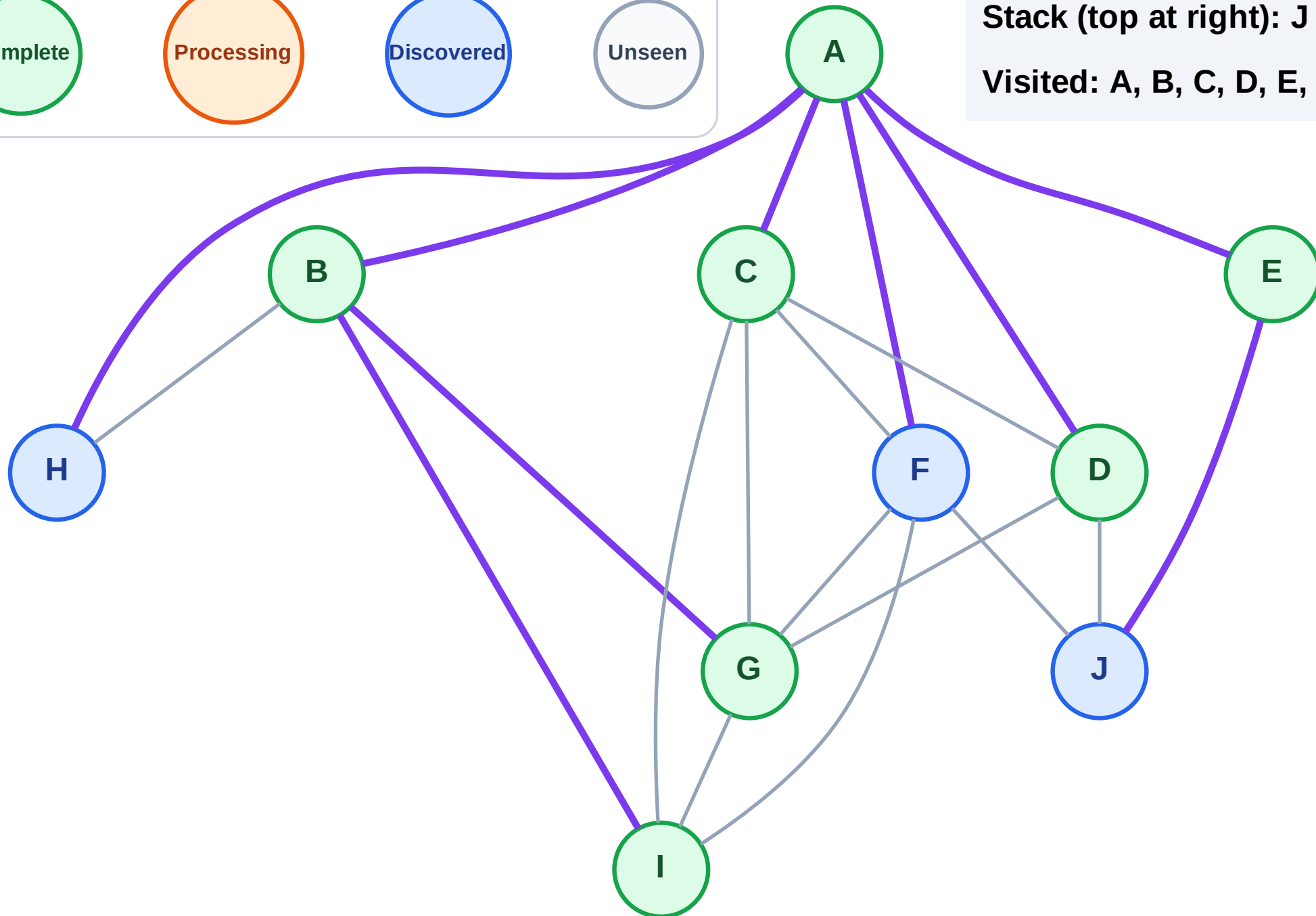
Step 15: No new neighbors from D

Legend



Stack (top at right): J | H | F

Visited: A, B, C, D, E, G, I



DFS Traversal

Step 16: Process node F

Legend

Complete

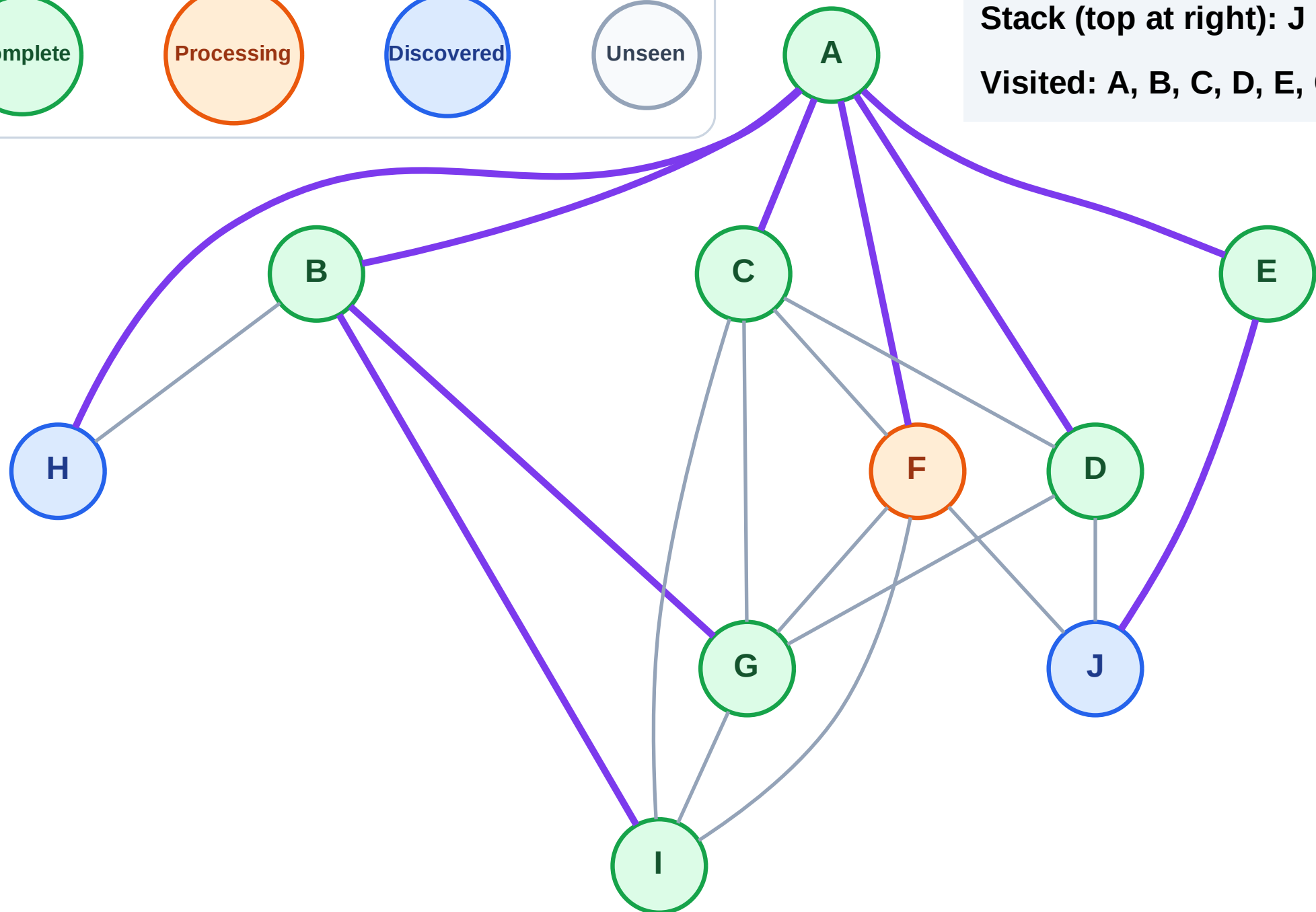
Processing

Discovered

Unseen

Stack (top at right): J | H

Visited: A, B, C, D, E, G, I



DFS Traversal

Step 17: No new neighbors from F

Legend

Complete

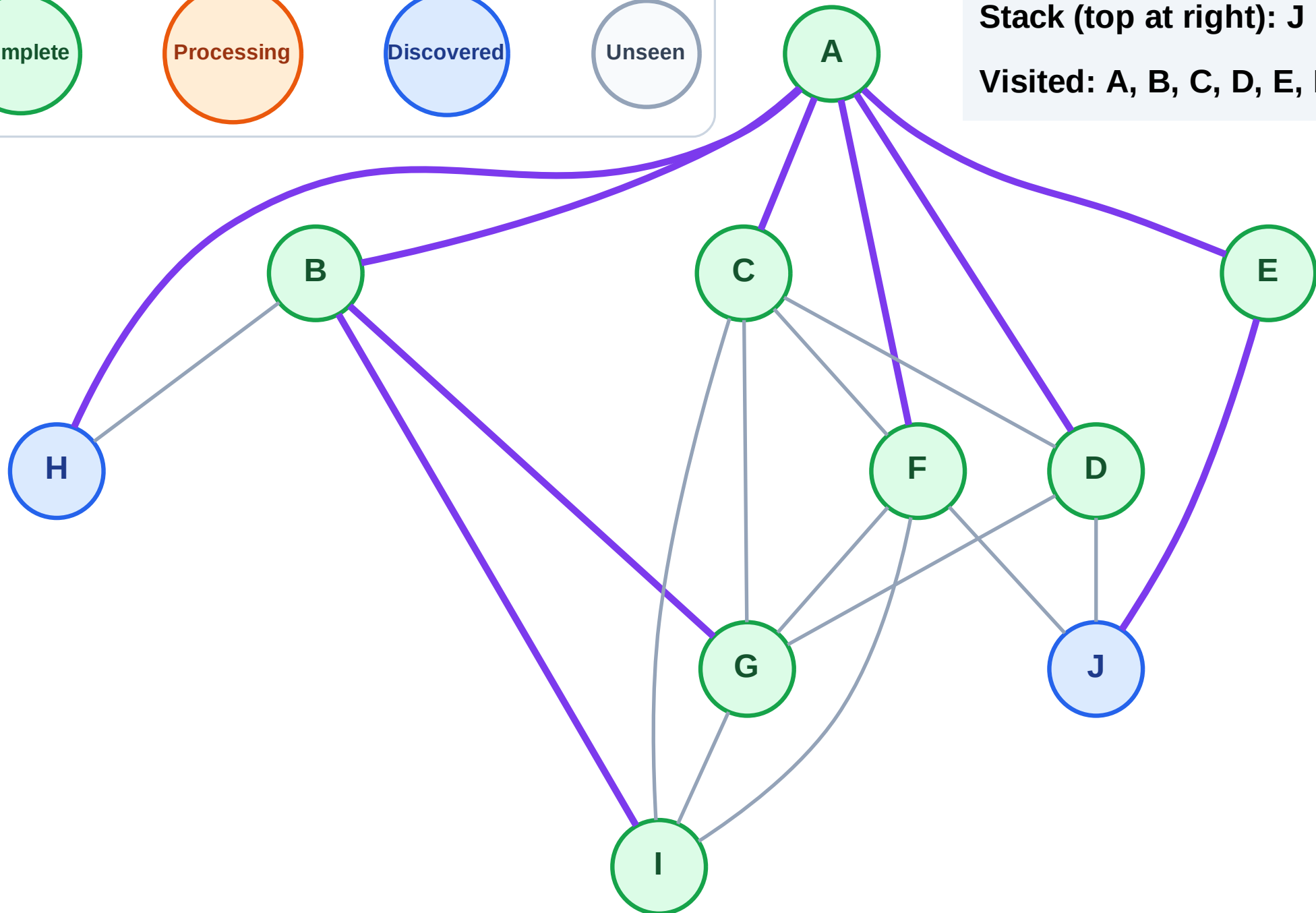
Processing

Discovered

Unseen

Stack (top at right): J | H

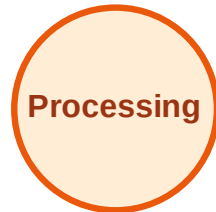
Visited: A, B, C, D, E, F, G, I



DFS Traversal

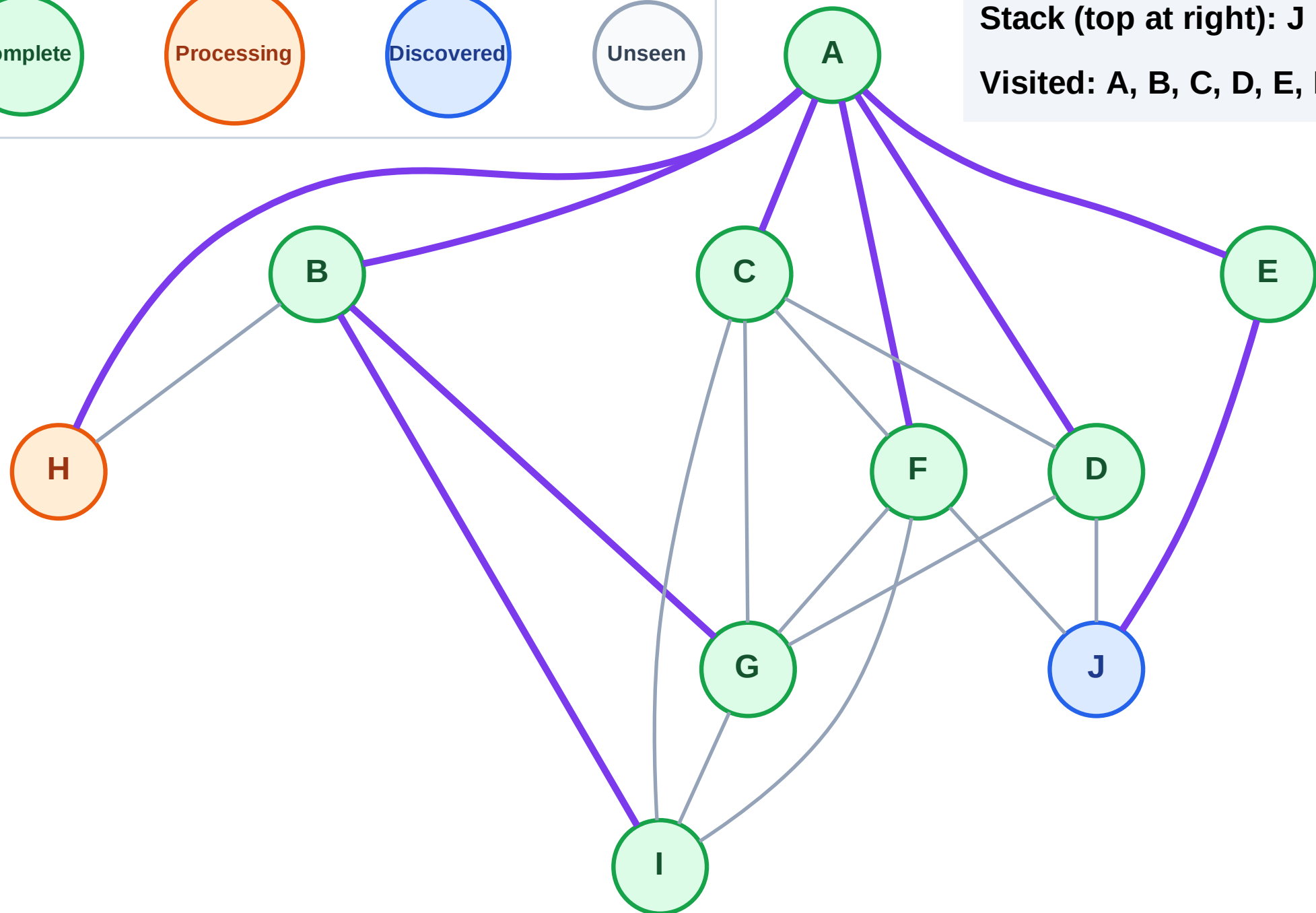
Step 18: Process node H

Legend



Stack (top at right): J

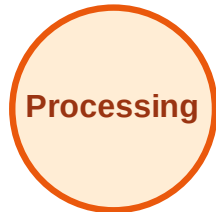
Visited: A, B, C, D, E, F, G, I



DFS Traversal

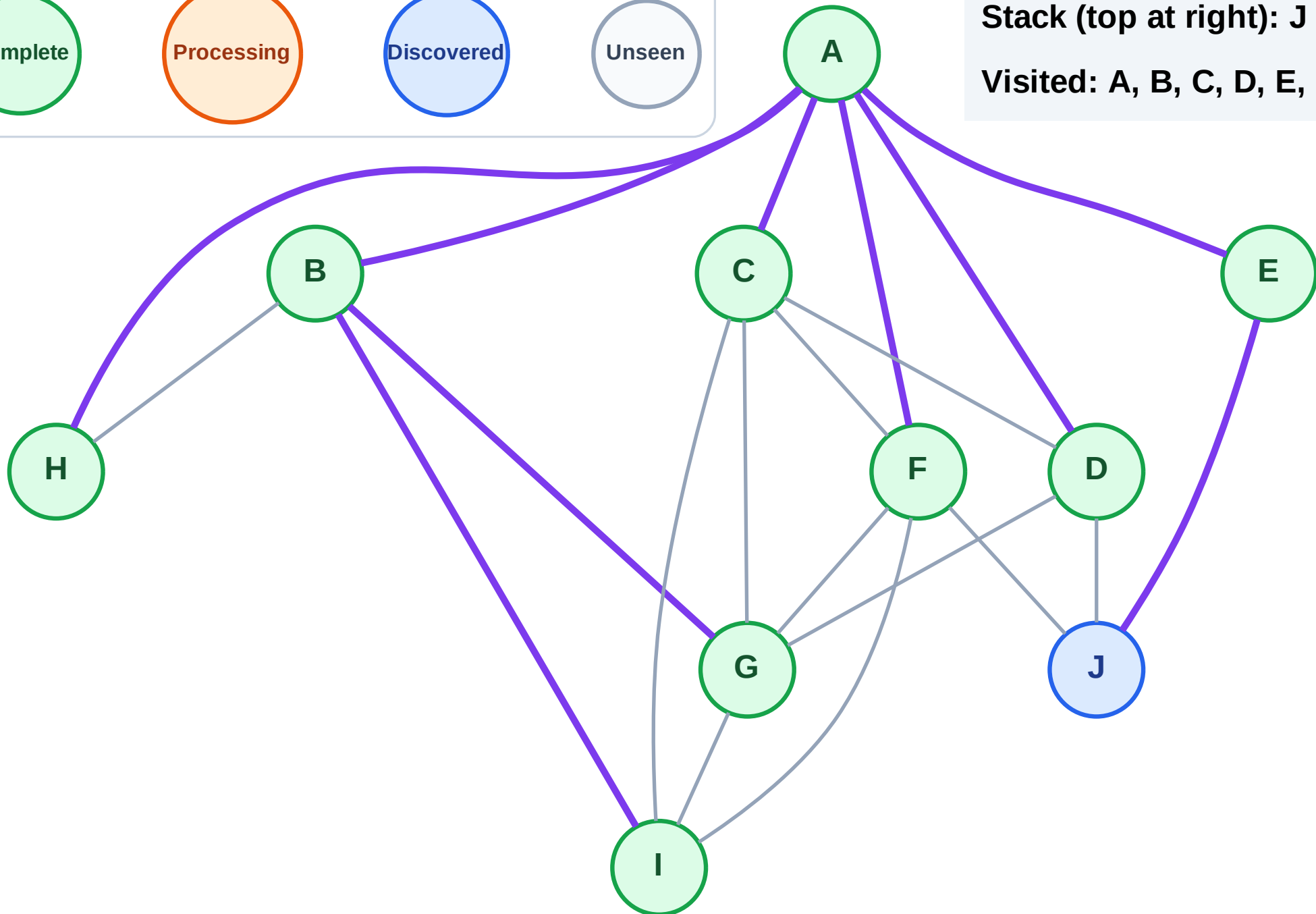
Step 19: No new neighbors from H

Legend



Stack (top at right): J

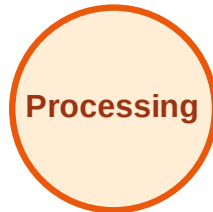
Visited: A, B, C, D, E, F, G, H, I



DFS Traversal

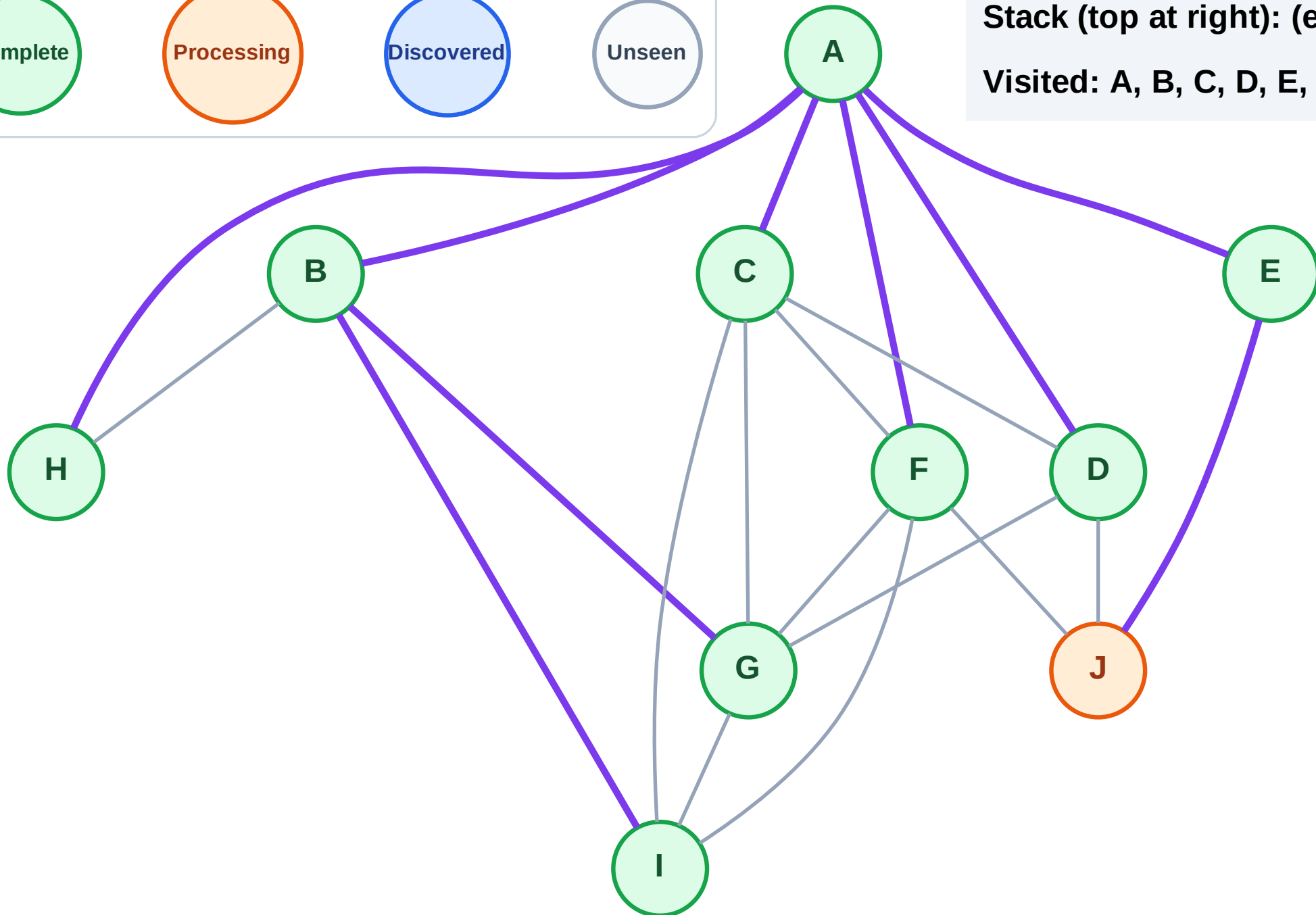
Step 20: Process node J

Legend



Stack (top at right): (empty)

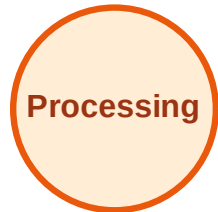
Visited: A, B, C, D, E, F, G, H, I



DFS Traversal

Step 21: No new neighbors from J

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

